

Selective Cinema

Blink-Rate Glasses and the Multiplexing of Narrative Experience

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Preface

This book concerns a simple device and a less simple consequence. The device is a pair of glasses whose lenses open and close in step with a projector, admitting some frames and rejecting others. The consequence is that one screen, shown to one room, can carry several films at once, and that the people in the room need not see the same one.

The idea did not begin as a proposal about cinema in general. It began as a detail of a fictional production: a science-fiction thriller called *Rarely Needed Protocols* and a satirical children's comedy called *Barely Needed Protocols*, written to be screened simultaneously, sharing a complementary soundtrack, and separated only by the shutter timing of the glasses each viewer wore. The same notes proposed that the glasses could be mechanical, like the shutter of an old projector, and that which film a viewer saw would depend on which frame the shutter started on. A separate set of notes proposed the inverse arrangement: a gallery that appears empty until glasses with the correct blink sequence reveal the work, and the possibility of patenting the sequences so that only licensed glasses could see. Both sets of notes are archived in public repositories with their commit histories, and Appendix G quotes them.

Those early notes already contained most of what this book develops: selection by timing rather than by content, the need for sound that serves more than one picture, the difference between choosing a version and being prevented from seeing one, and the uneasy fact that a mechanism for selection is also a mechanism for control. What they did not contain was an account of what kind of object a film becomes when it is designed from the start to be seen in several ways at once. Most of the book is an attempt to supply that account.

The argument proceeds from hardware to meaning. Part I treats the screen as a carrier of interleaved streams and asks what the optics can and cannot do. Parts II through V treat the film as a family of related realizations and ask how such a family is written, scored, shot, and edited. Parts VI through IX turn to the audience and to the institutions that decide which realizations exist and who receives them. The book's central claim, stated in the Introduction and developed in Chapter 17, is that collective viewing does not require that everyone see the same thing. It requires that different experiences remain synchronized closely enough to count as one occasion.

Introduction: One Screening, Several Films

Two people sit next to each other in a cinema. The lights go down, the film begins, and both of them look at the same screen for the next hour and a half. When they leave, one of them has seen a comedy and the other a drama. Their films began together, reached their principal turns together, and ended together. The same characters moved through the same rooms. But the lines were delivered differently, the scenes were cut to a different rhythm, and the last ten minutes resolved in opposite directions. Neither person looked at a phone, wore a headset, or watched a second screen. Each wore a pair of glasses.

The scenario seems to require something impossible, because it seems to require that one illuminated surface present two different images at the same moment. It does not. A projected film is not a continuous image. It is a rapid sequence of discrete frames, each shown for a fraction of a second, and the impression of continuous motion is produced by the visual system rather than by the screen. Once the image is understood as a sequence, the scenario becomes a matter of scheduling. If the projector alternates frames from two films, and each viewer's glasses admit only the frames of one of them, then two films occupy one screen. Each viewer sees a complete film at half the frame rate the projector runs at, and neither sees the other's.

What is and is not new

The optical principle is old and has been implemented several times. Laurens Hammond's Televue system, patented in November 1922 (US 1,435,520) [7] and first shown publicly at New York's Selwyn Theatre on December 27 of that year, gave each seat a rotary shutter synchronized to interlocked projectors alternating left-eye and right-eye images, so that each eye saw only its own frames and the audience saw depth. Televue is a direct ancestor of the apparatus, though not of the idea: its two streams were halves of one picture, not two stories. Modern stereoscopic cinema and home television have used electronic shutter glasses for the same purpose. Sony's SimulView, announced in August 2011 for the PlayStation 3D Display released later that year [14], turned the stereoscopic arrangement sideways: instead of sending different frames to a viewer's two eyes, it sent different full-screen images to two players, so that each saw only their own view of a split-screen game without the split. It is the closest commercial precedent for co-located viewers receiving different full-screen streams, though its purpose was multiplayer play rather than authored films. Research on what has been called temporal psychovisual modulation has explored displays that show one image to viewers wearing synchronized glasses and a different one to the unaided eye [15].

None of this is claimed here as a discovery. What the book adds is what those systems did not need. A game shown to two players does not need the two views to tell a coherent story together. A stereoscopic film does not need its left and right images to diverge in meaning. The present proposal requires both. It asks for several films designed as members of one family: synchronized in time, sharing characters and events, differing in treatment, and able to be seen in the same room without any of them being the real one and the others alternatives. The engineering is a starting point. The subject is the form.

Neighbors and differences

Several existing practices resemble the proposal in part, and the differences clarify what it is.

Stereoscopic cinema already sends different images to different eyes, but it recombines them into a single percept. The two streams are halves of one experience, not two experiences.

Captions and audio description add information to a common image without replacing it. A captioned viewer and an uncaptioned viewer see the same film with different supplements.

Alternate cuts, such as director's cuts, broadcast edits, and censored versions, are common, but they are distributed separately. They are watched at different times, in different places, by audiences who never share a room.

Branching film, in which the audience chooses what happens next, produces different experiences from one work, but only by interrupting it. *Kinoautomat*, shown at Expo 67 in Montreal, stopped at decision points and let the audience vote, so that the whole room followed the majority's branch [8]: a shared choice with one outcome for everyone. Streaming branching video such as *Black Mirror: Bandersnatch* gives each viewer or household its own path, but only by giving up the shared room [9]. Selective cinema lets each viewer take their own branch without stopping the film and without leaving the room.

Head-mounted displays can show every viewer something different, but they do so by replacing the shared screen with private ones. The room becomes a set of individuals who happen to be near each other.

Selective cinema, as the proposal will be called, keeps what each of these gives up. It keeps the common screen, the uninterrupted screening, and the shared room, and it still lets the audience see different films. The combination is what makes it more than a curiosity. It separates two things ordinary cinema has always bundled together: being present at the same event and receiving the same images.

The terms

A few terms need fixing at the start.

A *stream* is one sequence of frames within the projection, the frames a particular pair of glasses admits. A *realization* is the complete audiovisual experience a viewer receives by following one stream, with its sound. A *variant family* is the set of realizations designed together for synchronized presentation.

Blink-rate glasses are glasses whose lenses open and close on a schedule synchronized to the projection. The name refers to the schedule of the shutter, not to the wearer's own blinking. The glasses do not watch the viewer; they are instruments of timing. A schedule is specified

by two numbers, a period and a phase, and Chapter 2 shows that this pair, not a rate alone, is what selects a stream.

Selection is choosing a stream. *Concealment* is preventing someone from seeing a stream they have not chosen. The two are easily confused, and a large part of the book depends on keeping them apart. A mechanism that works well for selection may work poorly for concealment, and a system sold for concealment inherits obligations a system built for selection does not.

The argument

Part I treats the screen as a carrier. It asks how much temporal capacity a projector has, how many streams that capacity can support, what the glasses must do to separate them, and what the room sees when someone takes the glasses off. The answers impose costs that grow with every added stream, and they also reveal a design freedom that is easy to miss: the composite image seen without glasses can itself be designed.

Part II treats the film as a family of realizations. Its central notion is *switch admissibility*, the condition under which a viewer can move from one stream to another without the new stream presupposing something the viewer was never shown. Synchronization anchors, narrative rendezvous, and the accounting of time borrowed and repaid across streams all follow from that condition. The part then works through the principal kinds of variation: genre, explanatory depth, and endings.

Part III is devoted to sound, which is harder to multiplex than light. Air cannot be interleaved the way frames can. The part defends a solution present in the idea from the start: a single composed score that serves every stream at once, keeping the room acoustically shared while the pictures diverge.

Parts IV and V take up viewpoint, local realization, and the practical work of writing, producing, and editing a variant family. Part VI turns to the audience: what people share when they share a room but not a picture, what they say to each other afterward, and how and when they should be allowed to choose. Part VII considers accessibility, instruction, and translation, where selective streams could do real good and could also sort people without their consent.

Part VIII confronts the hardest questions. Someone must decide which versions exist and who can see them. The same glasses that let a viewer choose a comedy could let a distributor show different audiences different claims while everyone believes they saw one film. The same timing that selects a stream could be licensed so that only approved glasses can see at all. The part argues for disclosure, for glasses that record nothing, and for an account of the work's identity that locates it in its declared constraints rather than in any single sequence of frames. Part IX sets out how such a system should be evaluated.

The principal claim of the book can be stated now:

Collective viewing does not require that everyone see the same thing. It requires that different experiences remain synchronized closely enough to count as one occasion.

Everything that follows is an attempt to say what “closely enough” requires, what it costs, and who should decide.

Part I

The Screen as Carrier

Chapter 1

The Discrete Image

Cinema presents itself as continuous and is not. The screen of a film projector is dark for a substantial part of every second, and the image that is shown is a succession of stills. The continuity belongs to the viewer. This chapter makes that fact precise, because selective cinema is built entirely in the gaps it exposes.

1.1 Four kinds of frame

It is common to say that a film runs at twenty-four frames per second, and the statement hides several different quantities.

The *capture rate* is how many distinct images the camera records per second. For most theatrical film it is twenty-four.

The *encoded rate* is how many distinct images the distributed file or print contains per second. It usually equals the capture rate, but it need not: frames can be interpolated, dropped, or synthesized.

The *flash rate* is how many times per second the projector actually illuminates the screen. A mechanical film projector with a two-bladed shutter shows each frame twice, so twenty-four distinct images produce forty-eight flashes. A three-bladed shutter produces seventy-two. Digital projectors are not bound to a mechanical shutter, but many still repeat frames, and stereoscopic digital exhibition has commonly used a “triple flash” arrangement in which each of the two eyes’ images is shown three times per frame, giving $24 \times 2 \times 3 = 144$ flashes per second.

The *perceived rate* is not a number at all. It is the viewer’s impression of motion and steadiness, which depends on the flash rate, the brightness of the image, the speed of the movement depicted, where on the retina the image falls, and whether the eyes are tracking something.

These four quantities are separable, and the separability is the resource. The flash rate is fixed by the projector. What is shown in each flash is not. A projector running at 144 flashes per second is, in effect, a device with 144 slots per second, and nothing about the device requires those slots to carry repetitions of one image rather than images from several films.

1.2 Display bandwidth

Call the flash rate of a display its *refresh rate* R , and each flash a *slot*. A conventional film uses its slots to show one sequence, with repetition to reach a flash rate high enough to avoid visible flicker. Write the displayed sequence as

$$F = (f_1, f_2, \dots, f_n), \quad (1.1)$$

where each f_i is the image shown in slot i . Repeated flashes of one frame appear as consecutive equal entries.

A two-stream projection assigns alternate slots to two films,

$$P = (a_1, b_1, a_2, b_2, \dots, a_n, b_n), \quad (1.2)$$

and more generally an m -stream projection is any assignment of slots to streams. The relevant resource is therefore not the narrative frame rate of any one film but the *temporal display bandwidth* R : the number of slots per second available to be divided among whatever is being shown. Ordinary cinema spends most of that bandwidth on repetition. Stereoscopic cinema spends it on two eyes. Selective cinema proposes to spend it on several films.

Definition 1.1 (Display schedule). A *display schedule* is a function σ from slots to streams, $\sigma(i) \in \{1, \dots, m\}$, together with the image $P(i)$ shown in each slot. Stream k is the subsequence of images in slots with $\sigma(i) = k$.

The definition allows schedules that are not simple alternation. A stream may receive more slots than another, slots may be grouped in blocks, and, as Chapter 2 shows, a slot may belong to more than one stream if the streams agree on what it should show.

1.3 Two constraints from the eye

Two properties of human vision limit how finely the bandwidth can be divided.

The first is *flicker*. A light that switches on and off appears steady only above a certain rate, the critical flicker fusion frequency. That frequency is not a constant. It rises with brightness, rises in peripheral vision, and varies between individuals. A dim image viewed centrally may look steady at around fifty flashes per second; a bright image seen at the edge of vision may flicker visibly at seventy or more. What matters for selective cinema is the flash rate each *viewer* receives, not the rate of the projector. If the glasses admit one slot in three from a 144-slot projector, the viewer receives 48 flashes per second, which is the flash rate of a two-bladed film projector and is marginal for a bright image.

The second is *motion rendition*. Even when flicker is invisible, a moving object sampled too sparsely looks jerky or doubled, especially when the eyes follow it across the screen. Motion rendition depends on the number of distinct images per second, not on the number of flashes: repeating a frame three times removes flicker but does nothing for judder. Selective cinema therefore has to satisfy two budgets at once. Each stream needs enough flashes to avoid flicker, and enough distinct images to render motion acceptably.

The two budgets are why the division of bandwidth is not a matter of simple arithmetic, and why the next chapters treat capacity with some care. A stream of 24 distinct images per second can be shown with one flash each or three; the second costs three times as many slots

and looks steadier. A stream showing a slow conversation and a stream showing a chase make different demands, and a schedule that ignores the difference wastes slots on the first and starves the second.

1.4 What the glasses do to time

Glasses that admit some slots and block others change the viewer's sampling of the screen. They do not change the screen. The consequence is easy to state and important for everything that follows: the glasses can only remove light, never add it. A viewer wearing them sees at most the fraction of the projector's output that falls in admitted slots, and sees it through lenses that are never perfectly transparent.

A second consequence concerns the moments between slots. No real shutter opens and closes instantly, and no real display switches instantly between images. At each boundary there is a brief interval in which the glasses are partly open while the screen is changing, or fully open while the previous image lingers. What leaks through in those intervals is the subject of Chapter 2, and it matters more in selective cinema than in stereoscopy. A stereoscopic leak shows one eye a trace of the other eye's view of the same scene, and the result is a faint ghost. A selective leak shows a viewer a trace of a different film.

1.5 Summary

Cinema is a schedule of discrete flashes, and the schedule has more capacity than one film uses. That capacity, the temporal display bandwidth, is what selective cinema divides. The division is constrained by flicker, which limits how few flashes a viewer can receive, and by motion rendition, which limits how few distinct images a stream can contain. The glasses that perform the division can only subtract, and their imperfect timing lets fragments of other streams through. The next chapter specifies the glasses.

Chapter 2

Period and Phase

This chapter specifies the glasses. It begins with a correction of terminology, moves to the one idea on which the whole apparatus depends, and ends with the ways the apparatus fails.

2.1 What “blink rate” means

The name invites a misreading. It suggests glasses that respond to the wearer’s own blinking, perhaps by detecting how often the eyelids close and switching streams accordingly. Nothing of the kind is intended. The blinking belongs to the lenses. Each lens is a shutter that opens and closes on a schedule synchronized to the projector, and the “rate” is the rate of that schedule. The glasses do not observe the wearer at all. This is not a minor clarification. A device that measured the wearer’s eyes would be a sensor pointed at a person, with everything that implies about data and consent. A device that only times its own shutter is an optical filter, and it can be built so that it has no means of measuring anything.

2.2 Rate is not enough

Suppose the projector runs at 60 slots per second and alternates two films, one in the odd slots and one in the even slots. A viewer who wants either film needs glasses that open 30 times per second. Both viewers’ glasses therefore have the same rate. What distinguishes them is where in the cycle the shutter opens: on the first slot or on the second. The film a viewer sees is determined not by how often the shutter opens but by which slot it starts on.

This is the essential point, and it generalizes. Number the slots $i = 0, 1, 2, \dots$. A periodic shutter schedule is fixed by two integers: a *period* $p \geq 1$, the number of slots in one cycle, and a *phase* $\varphi \in \{0, \dots, p-1\}$, the slot within the cycle at which the shutter opens.

Definition 2.1 (Periodic gate). The periodic gate with period p and phase φ is the function

$$g_{p,\varphi}(i) = \begin{cases} 1 & \text{if } i \equiv \varphi \pmod{p}, \\ 0 & \text{otherwise.} \end{cases} \quad (2.1)$$

Its *admitted set* is $S_{p,\varphi} = \{i : g_{p,\varphi}(i) = 1\}$.

A viewer wearing glasses set to (p, φ) sees exactly the images in $S_{p,\varphi}$. At a projector refresh rate R , the viewer receives R/p flashes per second regardless of phase. The phase changes nothing about how much the viewer sees, only what.

Proposition 2.1. For fixed p , the admitted sets $S_{p,0}, \dots, S_{p,p-1}$ are pairwise disjoint and together contain every slot.

Proof. Every integer has exactly one residue modulo p . □

A projector with period- p scheduling therefore carries exactly p non-overlapping streams, one per phase, each at rate R/p . At 144 slots per second, period 2 gives two streams of 72 flashes per second, period 3 gives three streams of 48, and period 6 gives six streams of 24.

2.3 Nested periods

Periods need not be the same for every viewer. If one period divides another, the coarser schedule contains the finer ones.

Proposition 2.2. If p divides q , then

$$S_{p,\varphi} = \bigcup_{\substack{0 \leq \psi < q \\ \psi \equiv \varphi \pmod{p}}} S_{q,\psi}. \quad (2.2)$$

Proof. A slot i lies in $S_{p,\varphi}$ exactly when $i \equiv \varphi$ modulo p . Its residue ψ modulo q then satisfies $\psi \equiv \varphi$ modulo p , because p divides q ; conversely any such ψ gives $i \equiv \varphi$ modulo p . □

The proposition has a direct use. Suppose a projector runs four streams at period 4. A viewer at period 2 and phase 0 admits the slots of streams 0 and 2 together. If those two streams were designed as, for example, the comic and dramatic versions of the same scene, the period-2 viewer receives both interleaved, at twice the flash rate and twice the brightness of either, and sees their temporal average. Whether that average is meaningful depends on whether it was designed to be, which is the subject of Chapter 4. Nesting turns the set of available schedules into a hierarchy: fine settings select individual streams, and coarse settings select designed combinations of them.

2.4 Shared slots

Periodic gates of one period partition the slots. Nothing requires the streams of a variant family to be disjoint in this way. Two realizations of the same film share a great deal: establishing shots, action that both versions depict identically, the credits. During such passages the streams do not differ, and there is no reason to spend separate slots on them.

The general case is best described by the admitted sets directly. Each stream k is assigned a set S_k of slots its glasses open for, and a slot may belong to several streams if the image in it is correct for all of them. For example, with period 3, let phase 0 be common, phase 1 belong to realization A , and phase 2 belong to realization B . Viewers of A admit phases 0 and 1; viewers of B admit phases 0 and 2. Each receives two slots in three rather than one in two. The common slot must carry an image that is right for both, which it can do whenever the two realizations coincide in that moment, and which constrains the design whenever they do not.

2.5 Adaptive gating

The shared-slot idea points to something more useful. The streams of a variant family diverge in some passages and coincide in others. When they coincide, every slot shows an image correct for every realization, and the glasses have no reason to close at all. If the synchronization signal carries not only the slot clock but also a flag indicating whether the streams currently coincide, the glasses can open fully during common passages and gate only during divergent ones. Coincidence must be judged on what the display emits, not on stored values: two streams coincide in a slot when their displayed images agree within a declared perceptual tolerance. Identical code values can differ on screen if the streams are graded differently, and different code values can be indistinguishable once displayed, so a test on stored data would be both too strict and too loose.

The effect is that the costs of multiplexing, in brightness and in flash rate, are paid only while the films actually differ. A comedy and a drama that share half their footage cost half as much light as two unrelated films. It also means that a viewer without glasses sees the correct image during common passages, because every slot shows the same thing. The unaided view degrades only where the family diverges, a fact Chapter 4 develops.

Adaptive gating requires the projector to know, slot by slot, which streams coincide. That knowledge belongs in the exhibition metadata and is produced during editing (Chapter 16). It also creates a visible cue: a viewer can notice when the lenses stop darkening and infer that the streams have converged. Whether that cue should be suppressed, for instance to protect a surprise, is a design choice rather than a technical necessity.

2.6 Synchronization

The glasses must know where the projector is in its cycle. The usual arrangement is a beacon, infrared or radio, that transmits a timing reference, and a small oscillator in each pair of glasses that locks to it, so that brief interruptions of the signal do not immediately lose the phase. The beacon can also carry the adaptive-gating flag and a schedule identifier.

The phase a viewer selects is, in the simplest arrangement, a dial on the glasses. It is worth noting how little else the glasses need. A shutter, a timing circuit locked to the beacon, and a dial are sufficient. There is no need for a camera, an eye tracker, a network connection, storage, or an account. The simplest possible implementation is close to mechanical, a rotating or oscillating shutter driven in step with the projector, which is what the Teleview system of 1922 used for stereoscopy. Glasses built this way can be shown, by inspection, to be incapable of recording what their wearer sees or measuring how their wearer responds. That is an architectural guarantee, stronger than any policy, and Part VIII argues that it should be treated as a requirement rather than a limitation.

2.7 Leakage

The ideal model is that a viewer on stream k sees exactly the images in S_k . Write $P(t)$ for the image on the screen at time t and $g_k(t)$ for the ideal gate of stream k . The ideal received signal is

$$V_k(t) = g_k(t) P(t). \quad (2.3)$$

Real shutters take time to open and close, are not perfectly opaque when closed or perfectly clear when open, and may drift slightly out of phase. Real displays take time to change from one image to the next, and some leave a trace of the previous image. A more honest model is

$$V_k(t) = \tau_k(t) P_k(t) + \sum_{j \neq k} \epsilon_{kj}(t) P_j(t), \quad (2.4)$$

where $P_j(t)$ is stream j 's image, τ_k is the actual transmission of the glasses for their own stream, and ϵ_{kj} is the leakage of stream j into the view of stream k .

At 144 slots per second each slot lasts under seven milliseconds, and common liquid-crystal shutters take a non-negligible fraction of that to change state. The standard remedy is to insert a short dark interval between slots, blanking the screen while the shutters switch. This reduces leakage and costs light, because the dark interval is time in which nothing is shown. The trade between leakage and brightness is the first engineering choice any selective system has to make, and it has no free solution.

Leakage has two measures, and they come apart. The first is optical: how much light from other streams reaches the eye. The second is informational: whether that light carries anything recognizable. A faint, blurred trace of a different image is optically measurable and may be informationally empty. A brief, clear glimpse of a face, a word, or a final scene may be optically small and informationally decisive. Stereoscopic cinema could tolerate leakage because both streams showed the same scene. Selective cinema cannot assume that, and the acceptable level of leakage depends on what the leaked streams contain. Chapter 18 develops this into a graded standard tied to narrative consequence.

2.8 Phase is not a secret

A final property of the apparatus sets up the political questions of Part VIII. In the scheme described here, the phase of each stream is not hidden. Anyone with adjustable glasses can turn the dial and see any stream. The system selects; it does not conceal.

Concealment would need something more. One possibility is a schedule that is not periodic: slots assigned to streams by a pseudorandom sequence generated from a key, so that only glasses holding the key open at the right moments. Such a system is closer to encryption than to selection, and it changes what the glasses are. They now carry a secret, need a way to receive it, and can be licensed, revoked, or restricted to approved hardware. The early notes behind this book imagined exactly this, patented blink sequences and “blink-rate protectors” that let only authorized glasses see a work, and the idea is both technically plausible and the point at which a viewing device becomes an instrument of control. Chapter 23 examines it. For now the distinction can be fixed in one sentence: periodic gating makes a film selectable, and keyed gating makes it restricted, and the two should not be sold under the same description.

Chapter 3

Capacity and Luminance

How many films can one screen carry? The question has a simple answer and a correct one, and the difference between them is where the design work lies.

3.1 The simple bound

If a display has refresh rate R and every stream needs at least $r_{\min}$ flashes per second to look steady, then evenly interleaved streams, each receiving R/m flashes, satisfy the requirement only if

$$m \leq \left\lfloor \frac{R}{r_{\min}} \right\rfloor. \quad (3.1)$$

With $R = 144$ and a flicker floor of 48 flashes per second, three streams fit. With a floor of 72, two fit. The bound is useful as a first estimate and wrong in three ways: it ignores the time lost to switching, it treats flicker as the only visual requirement, and it assumes every stream needs the same share at every moment.

3.2 The correct budget

Chapter 1 separated two visual requirements. A stream needs enough flashes to avoid flicker and enough distinct images to render motion. Suppose stream k shows d_k distinct images per second and repeats each n_k times, so that it receives $d_k n_k$ flashes per second. Suppose also that a fraction b of the display's time is spent blanked while the shutters switch, as described in Chapter 2. Then a schedule is feasible only if

$$\sum_{k=1}^m d_k n_k \leq (1 - b) R, \quad d_k n_k \geq r_{\min}, \quad d_k \geq d_{\min}. \quad (3.2)$$

The first condition says the streams together cannot use more slots than exist. The second is the flicker floor, and the third the motion floor.

Two consequences follow immediately. First, blanking is not a small correction. If a fifth of the time is spent dark, a fifth of the capacity is gone. Second, the flicker floor applies to flashes and the motion floor to distinct images, so a stream can meet the first cheaply by repetition only if it meets the second already. A film captured at 24 images per second, shown once

each, receives 24 flashes, well below any acceptable flicker floor; it needs at least two flashes per image, and at bright levels three. A stream captured at 48 images per second, shown once each, meets both floors with the same number of slots that a 24-image stream shown twice would use, and renders motion better. High-frame-rate capture therefore does not cost selective cinema anything extra. It spends the same slots better.

3.3 Unequal and changing allocation

Nothing in the budget requires every stream to receive the same share or to keep its share throughout the film. A long dialogue scene in one realization and a chase in another make different demands on motion rendition. A schedule could shift slots toward the stream that needs them and back again, provided each stream stays above both floors at all times.

Allocation can also change with coincidence. Chapter 2 described adaptive gating, in which the glasses open fully while the streams show the same image. During such passages the multiplex costs nothing, and the full display is available to everyone. Capacity is therefore not a fixed property of a film. It is consumed only during divergence, and a variant family with long shared passages can afford more streams during its short divergent ones than the simple bound suggests.

The limit on this flexibility is continuity of appearance. A stream whose flash rate or brightness jumps when the allocation changes will look as if the projector faltered. Changes must be gradual, or hidden at cuts, and editing must plan them (Chapter 16).

3.4 Luminance

Each viewer's glasses are closed for most of the cycle, and closed glasses pass no light. If stream k is admitted for a fraction δ_k of the display's time, the glasses transmit a fraction T of the light when open, and the display produces luminance L_P , then the viewer sees

$$L_k \approx \delta_k T L_P. \quad (3.3)$$

With two streams and ideal glasses, a viewer sees at most half the light; with three, a third. Real shutter glasses transmit well under all of the light even when open, and blanking reduces δ_k further. Stereoscopic exhibition, which already halves the light per eye and loses more to the glasses, is widely experienced as dim. A selective system that also carried stereoscopy would divide the light again.

Adaptive gating improves this in the same proportion as it improves capacity. If the streams coincide for a fraction c of the running time and diverge for the rest, the time-averaged luminance a viewer receives is approximately

$$\bar{L}_k \approx T L_P (c + (1 - c) \delta_k). \quad (3.4)$$

A two-stream family that coincides for half its length gives each viewer three quarters of the light of an ordinary screening, not half. The brightness then varies with the film, higher in shared passages and lower in divergent ones, which is itself a visible signal and must be managed like the allocation changes above.

3.5 The compensation trap

The obvious remedy for lost light is a brighter projector. It helps less than it seems to, for three reasons.

The first is flicker. The critical flicker fusion frequency rises with brightness. Raising the luminance to compensate for a low duty cycle makes each flash brighter, and brighter flashes need a higher flash rate to look steady. Compensation for light therefore tightens the flicker floor, which reduces the number of streams the display can carry. Light and capacity are not independent budgets. Spending on one draws down the other.

The second is the unaided view. Anyone in the room without glasses, or who lifts them for a moment, sees the full output of the projector. A screen bright enough to give glasses-wearers a comfortable image at a third of the duty cycle is three times too bright for everyone else.

The third is cost. More light means more power and more heat, and in large theaters light output is already close to what projectors can supply for ordinary two-dimensional exhibition.

The trap has one exit, and it is the same one as before. The cheapest light in selective cinema is the light not lost to gating, which means the most important economy is designing variant families that share as much as they can and diverge only where the divergence matters.

3.6 A realistic estimate

Taken together, the constraints suggest that a present-day cinema projector at 144 slots per second can carry two streams comfortably and three with care, if the films are captured at a high frame rate, blanking is kept short, and the family shares a substantial part of its running time. Displays with higher refresh rates, such as emissive LED screens, raise the ceiling, but brightness and flicker do not scale away. Each additional stream costs light that every viewer pays for.

This is a strong constraint on what selective cinema can promise. It rules out the image, suggested by some promotional descriptions of the idea, of a single screening offering a dozen independent films. It favors small families of closely related realizations over large menus of unrelated ones. That is not a defect in the proposal. It is a pressure toward the form the rest of the book argues for on other grounds: a few versions of one work, designed together, rather than many works sharing a wall.

Chapter 4

The Unfiltered Screen

Every selective screening has a realization that nobody chose. It is what the screen shows to an eye without glasses: to a child who has taken them off, to a viewer who cannot wear them, to an usher walking down the aisle. The first three chapters treated this view as a side effect, a problem of brightness and leakage. This chapter treats it as part of the work.

4.1 What the unaided eye sees

Above the flicker fusion rate, the visual system does not register individual flashes. It responds, approximately, to their average over a short integration window. A viewer without glasses therefore sees something close to the time-weighted mean of every stream on the screen. If stream j occupies a fraction δ_j of the slots, the unaided view is approximately

$$C(t) \approx \sum_{j=1}^m \delta_j P_j(t), \quad \sum_j \delta_j = 1, \quad (4.1)$$

where $P_j(t)$ is the image of stream j around time t . The average is of light, not of numbers. Here and throughout the chapter, images denote emitted luminance in linear light, not the gamma-encoded pixel values stored in files; averaging encoded values would predict the wrong composite. With two equal streams, C is the average of the two pictures: a double exposure.

During passages in which the streams coincide, every P_j is the same image and C equals it. The unaided viewer sees the film correctly. During divergent passages, the unaided viewer sees whatever the average of the divergent images happens to be. There are three ways to treat that average: tolerate it, suppress it, or design it.

4.2 Tolerating, suppressing, designing

To *tolerate* the composite is to accept whatever double exposure results. It is the default, and it is usually unpleasant: two unrelated scenes superimposed at reduced contrast, with edges and faces from each competing.

To *suppress* the composite is to make it deliberately uninformative, for example by arranging the streams so that their average is a uniform field. The unaided viewer sees gray, or darkness, or a neutral pattern, and knows only that something is being shown to others.

To *design* the composite is to arrange the streams so that their average is itself a meaningful image, a third realization available to everyone without glasses. The composite then becomes a member of the variant family, the one that belongs to the whole room.

4.3 The empty gallery

Suppression has an exact construction. Let G be a uniform gray image, and let A be any image whose luminance lies between zero and $2G$ at every point. Define a second image

$$B = 2G - A. \quad (4.2)$$

Shown alternately at equal share, A and B average to

$$\frac{1}{2}A + \frac{1}{2}B = G. \quad (4.3)$$

An unaided viewer sees a featureless gray field. The construction is exact only in linear light. In practice the images are linearized through the display's transfer function, the complement is computed there, the result is checked against the display's luminance range, and only then re-encoded for projection. The check matters, because B must be physically displayable:

$$0 \leq 2G - A \leq L_{\max} \quad \text{at every pixel}, \quad (4.4)$$

where $L_{\max}$ is the display's peak luminance. A viewer whose glasses admit only the A slots sees A . A viewer admitting only the B slots sees B , which is the complement of A about the gray level: light where A is dark, and dark where A is light.

This is the gallery that appears empty until the right glasses reveal the work, the idea in the early notes behind this book, and it is realizable with the apparatus of Chapter 2. It has three costs that the idea did not anticipate.

First, the contrast of the hidden image is limited by the gray level. The feasibility condition above confines A to the interval from $\max(0, 2G - L_{\max})$ to $2G$. A dark gray leaves the hidden image little room at the top; a gray near half the peak luminance gives it the most. The limited contrast is not a side effect of the design but a direct consequence of that inequality.

Second, the second stream is not free. B is determined by A . The empty gallery uses two streams' worth of display time to show one hidden image and its complement.

Third, and most important, the concealment works against eyes and not against cameras. A camera with a short exposure, synchronized by accident or design to the display, records individual slots rather than their average. A phone pointed at the gray wall can capture the hidden image. The early notes proposed, separately, a viewing room in which visitors are scanned to prove they carry no cameras or recorders and are then shown an individualized work alone. That proposal turns out to be the other half of the empty gallery. Temporal concealment hides a work from the unaided eye; only the absence of cameras keeps it hidden.

4.4 Designed composites in general

The construction generalizes. With m streams at shares $\delta_1, \dots, \delta_m$, and a target composite M , the streams must satisfy

$$\sum_{j=1}^m \delta_j P_j = M. \quad (4.5)$$

Any $m - 1$ of the streams can be chosen freely, within the display's range, and the last is then fixed:

$$P_m = \frac{1}{\delta_m} \left(M - \sum_{j < m} \delta_j P_j \right), \quad (4.6)$$

subject, as before, to computation in linear light and to the requirement that $0 \leq P_m \leq L_{\max}$ at every pixel.

Proposition 4.1. Designing the composite costs one stream's worth of freedom. A family of m streams with a prescribed unaided view has at most $m - 1$ independently chosen realizations, and the remaining one is constrained both in content and in range.

The proposition states a real trade. With two streams, a designed composite leaves one free realization and makes the other its complement, which is the empty gallery. With three, two realizations are free and the third is a correction term, which may or may not be watchable as a film in its own right. A family that wants every stream to be a good film and the composite to be a good film as well must find content for which all four conditions can hold at once. That is a compositional problem, not an optical one, and it is closer to the construction of an anamorphic painting or a piece of counterpoint than to projection engineering.

4.5 Where averaging fails

The unaided view is an average only approximately, and it fails in predictable places.

When the eyes move quickly across the screen, successive flashes land on different parts of the retina, and the average breaks into a scatter of individual frames, much as some single-chip digital projectors show brief color fringes during eye movements. A carefully designed gray field may show the hidden image for an instant during a saccade.

When objects move in one stream and not another, the average smears, and the smear can reveal the moving outline.

And the average holds only above the fusion rate. Near the flicker floor, and especially in peripheral vision, the alternation itself becomes visible as shimmer, and with it hints of the separate images.

A designed composite therefore degrades gracefully at best. It is reliable as an ambient view and unreliable as a guarantee of secrecy.

4.6 Nested viewers

Chapter 2 showed that glasses with a coarser period admit the union of several finer streams. A viewer so equipped sees a partial composite: the average of just those streams. This makes a family of designed composites available through the glasses themselves. In a four-stream family, a period-2 setting could show the average of streams 0 and 2 and another the average of 1 and 3, each designed as a coherent realization. A viewer could step from a fine setting to a coarse one and see two realizations fused into a third, then step back. Such hierarchies are demanding to compose, and they are the most distinctive artistic possibility the apparatus offers, because nothing like them exists in any other medium.

4.7 The composite and the children's cut

The analysis also settles a question the idea has often been used to illustrate: an adult film and a children's film on one screen, with glasses deciding which audience sees which. If one stream carries material unsuitable for children and a child removes the glasses, the child sees the composite, and the unsuitable material is in it, at reduced contrast but present. If the composite is suppressed to gray, the child sees gray during divergent passages, and the concealment holds against the eye while failing against any camera. In neither case do the glasses enforce anything. They let a viewer choose the children's film. They do not stop anyone from seeing the other.

This is the distinction the Introduction drew between selection and concealment, now in concrete form. The children's-cut example is useful for explaining the mechanism and misleading as a description of what the mechanism guarantees. The book therefore does not rest on it. Its founding case, taken up in Chapter 8, is different: a thriller and a comedy for children telling the same story, made to be watched side by side, with nothing in either that the other audience needs to be kept from. That arrangement asks the glasses only to select, which they do well, and it leaves the composite free to be designed as the room's own version of the film.

Part II

Variant Families

Chapter 5

The Film as a Family of Realizations

Part I established what the screen can carry. This part asks what it should carry. Two unrelated films of equal length can be interleaved, and the result is two films that happen to share a room. The proposal of this book is something else: a single work that exists as several related realizations, designed together and meant to be seen together. This chapter defines that object and distinguishes it from its weaker neighbors.

5.1 Realization and family

A *realization* is a complete audiovisual experience, one film a viewer can follow from beginning to end. A *variant family* is a set of realizations

$$\mathcal{F} = \{F_1, F_2, \dots, F_m\} \quad (5.1)$$

designed for synchronized presentation. Each member is whole. None is a supplement, a commentary track, or a set of optional scenes attached to another.

What makes a set of realizations a family rather than a program of short films is a shared *temporal scaffold*: a common clock against which every realization is laid out, and a set of moments at which the realizations are required to correspond. The scaffold is what lets several people in one room remain in the same occasion while watching different films, and it is what Part I's adaptive gating exploits when the realizations coincide on screen.

5.2 Degrees of relatedness

Realizations can be related more or less closely, and the degree determines what the family can do.

At the weakest level, the realizations merely have commensurate running times. They start and end together and share nothing else. Interleaving them is possible and pointless: the audience shares a room but not a work.

At the next level, the realizations share a structure: act boundaries, intermissions, major turns. They may tell different stories, but the stories move in step, and viewers can sense that they are at comparable points.

At a stronger level, the realizations share a world: the same characters, places, and events, differing in how the events are presented. A comic and a dramatic realization of one plot are related this way.

At the strongest level, the realizations are generated from a common model of the story, a scene graph of places, characters, and events on a single timeline, and differ only in the choices made in rendering it: which camera, which line reading, which explanation, which outcome at a branch. Every difference is a declared transformation of a common source.

The book concentrates on the last two levels, because only they preserve what the Introduction called a single occasion. Strangers watching unrelated films in one room are co-present. Viewers watching two realizations of one event are participating in one differentiated work.

5.3 Three kinds of distance

Two realizations can be near in one sense and far in another, and it helps to separate three measures of how far apart they are.

Image distance is how much of the time they show different pictures. Its complement is the coincidence fraction c of Chapter 3, and it governs light and flash-rate costs. Two realizations that share most of their footage are cheap to multiplex.

Fact distance is how much they differ in what they establish as having happened or being true: who did what, what was revealed, what the viewer is entitled to believe. It governs whether a viewer can move between them, which is the subject of Chapter 6.

Time distance is how far their events drift apart on the common clock: whether a scene in one realization runs longer than the corresponding scene in another, so that the two fall out of step. It governs synchronization and the editing of Chapter 16.

The three are analytically independent: each can in principle vary while the others are held fixed. Chapter 7 shows that real transformations tend to couple them. A comic and a dramatic version of the same scene, shot separately, have large image distance and almost no fact distance. Two versions that use identical footage but differ in one subtitle revealing the murderer have almost no image distance and decisive fact distance. A beginner's version that pauses to explain a step has small fact distance but grows in time distance with every explanation. Any account of what a family costs, or what it allows, has to say which distance it is talking about.

5.4 Primary and equal families

A family may or may not have a privileged member. In a *primary* family, one realization is the work in the conventional sense and the others are declared variants of it: an original with a children's adaptation, say, or a feature with an explanatory version for students. In an *equal* family, no member is privileged; the work is the family.

The distinction is a matter of declaration, not of optics, and it has consequences later. It determines what an archive must preserve, what a critic reviews, and whether a viewer who chose a variant can be said to have seen the film. Chapter 25 returns to it. Here it is enough to note that selective cinema does not force either answer. It makes the question unavoidable, where ordinary distribution lets it be deferred by releasing versions at different times to different audiences.

5.5 A family is not a menu

It is tempting to describe a variant family as a menu: a list of options from which each viewer picks. The description is misleading in two ways.

First, a menu's items are independent, and a family's are not. Each realization is constrained by the others through the shared scaffold. A scene in the comic realization cannot run long without the dramatic realization absorbing the difference somewhere, and a fact established in one cannot be contradicted in another without closing off the possibility of switching between them. The members of a family are written against each other.

Second, a menu offers a choice made once. A family offers a structure of choices across time. Whether a viewer can move from one realization to another, when, and in which direction depends on what each realization has shown so far. The family is therefore better understood as a set of paths through a common story, with a relation telling which paths can be joined to which at which moments. The next chapter defines that relation.

Chapter 6

Switch Admissibility

A viewer wearing adjustable glasses can turn the dial at any moment. Optically, nothing prevents it. Narratively, most moments are wrong. A viewer who switches from one realization to another may arrive in the middle of a scene whose meaning depends on something they never saw, or may be told something that contradicts what they saw a minute ago. This chapter defines when a switch works. The definition turns out to generate most of the structure a variant family needs: its anchors, its points of reconvergence, and the constraints its writers and editors must respect.

6.1 What a viewer carries

Fix the common clock of the family and write t for a moment on it. Each realization F_k presents a sequence of images and sounds; write $F_k^{\leq t}$ for what it has presented up to t , and $F_k^{> t}$ for what it will present afterward.

A viewer does not carry a realization. A viewer carries a *history*: what they have actually received. A viewer who watched realization a throughout has history $H = F_a^{\leq t}$. A viewer who watched a until time s and then switched to b has history

$$H(t) = F_a^{\leq s} \cdot F_b^{(s,t]}, \quad (6.1)$$

the first part of one realization followed by a stretch of another. Histories can contain several switches, and, if the viewer removed their glasses for a while, stretches of the composite of Chapter 4.

From a history, a viewer is entitled to certain beliefs about the story: who these people are, what has happened, what has been revealed. Write $\text{Est}(H)$ for the set of propositions a history *establishes*. This is not everything the viewer might guess, only what the film has presented as settled.

6.2 What a continuation needs

The future of a realization makes demands on the past. A scene in which a character is mourned presupposes that the viewer knows the character died. A scene in which an expert procedure is carried out without explanation presupposes that the viewer can follow it.

For each realization b and time t , write $\text{Pre}_b(t)$ for the set of propositions the continuation $F_b^{>t}$ relies on the viewer already holding.

The future also makes commitments. It will assert some things and take others for granted, including everything its own past established. Write $\text{Com}_b(t)$ for the propositions the continuation commits to, whether by asserting them later or by treating them as true. Every presupposition is a commitment, so $\text{Pre}_b(t) \subseteq \text{Com}_b(t)$.

6.3 The condition

Definition 6.1 (Switch admissibility). A viewer with history H can *admissibly* continue into realization b at time t , written $H \rightsquigarrow_t b$, when two conditions hold:

1. *coverage*: $\text{Pre}_b(t) \subseteq \text{Est}(H)$, so that everything the continuation relies on has been established for this viewer; and
2. *consistency*: $\text{Est}(H) \cup \text{Com}_b(t)$ is consistent, so that nothing the continuation commits to contradicts anything the viewer has been shown.

The two conditions fail differently. A coverage failure leaves the viewer lost: the new realization refers to things they never saw. A consistency failure leaves the viewer deceived: the new realization contradicts what they were told, and nothing in it acknowledges the contradiction. The first is confusing, and the second is worse, because the viewer may not notice.

Every realization should, of course, admit its own continuation. A realization in which $F_b^{\leq t} \rightsquigarrow_t b$ fails at some t is incoherent on its own terms, and the definition assumes this does not happen.

For the formal claims that follow, one simplification is useful. Treat propositions as literals, facts and their negations, treat consistency as the absence of any fact together with its negation, and let $\mathcal{R}$ be the set of literals that any continuation in the family depends on, closed under negation. The simplification is crude as a theory of narrative meaning and sufficient for the structural points.

6.4 Four consequences

Admissibility is not symmetric

That a viewer on a can move to b at time t says nothing about the reverse. Consider a family with an expert realization and a beginner realization of an instructional film. Treat each realization's declared prerequisites as established from its first frame: the expert realization takes its background as given from the start, while the beginner realization establishes the same background gradually by explaining it. At a moment when both are about to show the same step, the beginner continuation presupposes only what it has explained so far, and a viewer coming from the expert realization holds all of that. The switch from expert to beginner satisfies coverage. The switch from beginner to expert may not: the expert continuation presupposes the whole background, and the beginner realization has not yet supplied all of it. (Counting declared prerequisites as established is a presumption about viewers, not something the film has shown; Chapter 9 examines it.) The relation at each moment is a directed graph on the realizations, and its edges generally point one way.

Admissibility belongs to histories, not to pairs of realizations

It is tempting to tabulate, for each moment, which realizations can be switched between, and to treat the table as the whole story. The table is not enough.

Proposition 6.1. The admissibility of a path with several switches is not determined by the admissibility of each switch computed from single-realization histories.

Proof. Let the continuation of realization a after time s_2 presuppose a literal p . Let a establish p only during the interval $(s_1, s_2]$, and let b establish p only before s_1 . A viewer who watched only a can switch to b at s_1 , other things being equal. A viewer who watched only b holds p at s_2 , so the switch from b back to a at s_2 satisfies coverage. But a viewer who watched a until s_1 and b from s_1 to s_2 saw neither establishment of p , and the return to a at s_2 fails coverage. Each switch passes the single-realization test and the path fails. $\square$

Remark. The failure has a simple source. Each switch changes the history against which the next switch must be judged. Compatibility checked pairwise, between whole realizations, does not compose, because a viewer who has switched is no longer carrying any one realization. This is the reason selective cinema needs a theory of histories rather than a table of compatible cuts.

The same thing can happen with consistency: a literal picked up in one realization's segment can conflict with a third realization that neither of the others conflicted with.

A viewer who moves between realizations carries a mixture, and the mixture can lack what a realization needs, or conflict with it, even when neither of its parts did. Admissibility is therefore a relation between a viewer's entire history and a continuation. Any system that permits repeated switching has to reason about histories, or restrict switching to moments where histories do not matter.

Rendezvous make histories not matter

There are moments at which every realization has established the same things, as far as any continuation cares.

Definition 6.2 (Structural rendezvous). A time t is a *structural rendezvous* of the family when, for all realizations a and b ,

$$\text{Est}(F_a^{\leq t}) \cap \mathcal{R} = \text{Est}(F_b^{\leq t}) \cap \mathcal{R}. \quad (6.2)$$

Proposition 6.2. At a structural rendezvous, every single-realization history admits a switch into every realization.

Proof. Let t be a structural rendezvous and a, b any realizations. Coverage: $\text{Pre}_b(t)$ lies in $\mathcal{R}$ and in $\text{Est}(F_b^{\leq t})$, because b is coherent, hence in $\text{Est}(F_a^{\leq t})$. Consistency: a conflict would pair some literal $\ell \in \text{Com}_b(t) \subseteq \mathcal{R}$ with its negation in $\text{Est}(F_a^{\leq t})$. Since $\mathcal{R}$ is closed under negation, $\neg\ell$ lies in $\text{Est}(F_a^{\leq t}) \cap \mathcal{R} = \text{Est}(F_b^{\leq t}) \cap \mathcal{R}$, so b would establish $\neg\ell$ and commit to ℓ , contradicting its coherence. $\square$

A structural rendezvous is where the family's realizations, however different their surfaces, agree on everything that will matter. It is the natural place to allow switching, the natural place

for an intermission, and the natural definition of what the prospectus earlier called a synchronization anchor. The proposition covers single-realization histories only. A viewer who has already switched may carry literals no realization established on its own, and a rendezvous guarantees nothing about them unless the family also agrees on those. A family that wants unrestricted switching must therefore keep its divergences from establishing anything relevant that another realization could later contradict, which is a strong demand on writers.

A weaker notion covers viewers who have already switched.

Definition 6.3 (Viewer-relative rendezvous). A time t is a *rendezvous for the history H* when $H \rightsquigarrow_t b$ for every realization b .

A structural rendezvous is a rendezvous for every single-realization history. A viewer-relative rendezvous may exist for a particular mixed history even where no structural rendezvous does, and may fail for it at a structural one. Keeping the two notions apart prevents “rendezvous” from becoming either too restrictive, if only the structural kind is recognized, or falsely universal, if the structural guarantee is assumed to cover everyone. Previously switched viewers rejoin at viewer-relative rendezvous.

Time drift is absorbed, not excused

Realizations can drift apart on the common clock: one lingers on a scene the other passes quickly. The definition does not need a separate rule for this. A switch at clock time t lands wherever realization b is at t , and $\text{Pre}_b(t)$ and $\text{Com}_b(t)$ are computed there. If b is behind, the viewer may see again something they have seen, which is harmless. If b is ahead, its continuation may presuppose events the viewer has not yet reached, and coverage fails. Drift therefore shows up as admissibility failure, and the family’s clock discipline, the accounting of time borrowed and repaid across realizations in Chapter 16, is in part a discipline for keeping rendezvous possible.

6.5 The switch system

It would be convenient to summarize the relation, for designers and for projection metadata, as a sequence of directed graphs on the realizations, with an edge from a to b at time t whenever $F_a^{\leq t} \rightsquigarrow_t b$. Call this the *realization graph $\mathcal{G}_t$* . It is complete at a structural rendezvous, thin between rendezvous as the realizations establish different things, and keeps one-way edges like the expert and beginner example. It is a useful picture of the family’s shape.

It is not a correct basis for deciding switches. Its edges are computed from single-realization histories, and the proposition above shows that a path along its edges can fail. A memoryless graph on realizations reproduces exactly the pairwise error the counterexample exposes. The correct object must carry some summary of what the viewer has received. Under the literal simplification, the summary can be made exact.

Proposition 6.3. Whether a history H admits a switch into any realization at any later time depends on H only through the pair

$$q = \langle k, \text{Est}(H) \cap \mathcal{R} \rangle, \quad (6.3)$$

where k is the realization the viewer is currently watching.

Proof. Presuppositions and commitments of every continuation lie in $\mathcal{R}$, and $\mathcal{R}$ is closed under negation, so both coverage and consistency depend only on $\text{Est}(H) \cap \mathcal{R}$. The later evolution of that set, while the viewer stays on k or switches again, depends only on the current realization and on what it goes on to establish. $\square$

The *switch system* of the family is therefore a transition system whose states at time t are the reachable pairs q_t , and whose transitions are the admissible switches between states. Distinct histories with the same state are equivalent for all future switching, so the system stores equivalence classes of histories rather than the histories themselves. The number of reachable states is bounded by the number of realizations times the number of subsets of $\mathcal{R}$, but in practice it is far smaller, because most combinations of established facts cannot be produced by any sequence of switches. If tone or other constraints outside the literal model are tracked, the state grows to include them, but the principle is unchanged: the state must remember as much of the past as future admissibility depends on, and no less.

This is the right object for exhibition. The projector's metadata can carry the reachable states and their transitions, and glasses that keep a small record of their own state, updated only by the switches their wearer makes, can use it without recording anything about the wearer beyond the position of their own dial. What to do with the information is an artistic and ethical decision, not a technical one. The glasses could allow only admissible switches, locking the dial otherwise. They could allow any switch and signal whether it is admissible. They could allow switching only at structural rendezvous, where no state needs to be kept at all. Chapter 19 considers these options and argues that the artist may legitimately constrain choice. The point here is that constraint can now be principled: not a blanket prohibition, but a restriction to switches that keep the viewer's history coherent.

6.6 What the definition leaves out

The definition concerns what a viewer is entitled to believe. It says nothing about feeling. A switch from a tragic realization to a comic one at the moment of a death may satisfy coverage and consistency perfectly and still be grotesque. Tone has continuity conditions of its own, and they are real, but they are matters of judgment rather than of entailment, and the book does not attempt to formalize them. The switch system marks where a move is coherent. Whether it is bearable remains for the makers, and the viewer, to decide.

The definition also treats establishment as settled, when real films often establish things provisionally, mislead deliberately, and correct themselves later. A mystery's false suspect is established and then withdrawn. The framework accommodates this, because *Est* can be defined over what the film presents as settled at each moment, and withdrawals are simply literals that change. It does make one prediction: switching during a deliberate deception is hazardous, because the viewer may carry the deception into a realization that has already withdrawn it, or has never made it.

6.7 Selection as continuity

The chapter's result can be put simply. Choosing a realization is not like choosing an item from a menu, because the choice is made by someone who already carries a past. The rele-

vant question is never only which film a viewer wants, but which films their history can be continued into. Selection in a variant family is a continuity relation between a history and a future, and the family's structure, its rendezvous, its drift, its one-way edges, is the shape of that relation over time.

Chapter 7

The Grammar of Controlled Divergence

The members of a variant family differ from one another in particular ways, and the ways are not all alike. Some differences cost light and nothing else. Some close off switching for the rest of the film. Some make the realizations drift apart in time. This chapter classifies the principal kinds of difference by their effects on the three distances of Chapter 5 and on the switch system of Chapter 6. The classification is meant as a working grammar for writers: a way of knowing, before a scene is written, what a given kind of divergence will cost the family.

7.1 Tonal transformation

A tonal transformation changes how events feel without changing which events occur: comic or dramatic performance, different music, different pacing of reactions, a different emphasis in framing. The comic and dramatic realizations of one plot are the paradigm.

Image distance is usually large, because tone lives in performance and cutting, and those are on screen. Fact distance is small by design: the same things happen and the same things are revealed. The realization graph therefore stays close to complete, and tonal families are the easiest to move between, subject to the limit noted at the end of Chapter 6: a switch can be coherent and still be jarring. Time distance is the hidden cost. Comedy and drama keep time differently, and Chapter 8 shows that this is where tonal families are hardest to build.

7.2 Expository transformation

An expository transformation changes how much the film explains: background supplied or assumed, terms defined or used, steps shown or skipped. Beginner and expert realizations are the paradigm.

Fact distance is asymmetric. The expert realization takes its background as given from the start; the beginner realization establishes the same background as it goes. Until the beginner realization has caught up, expert viewers can move into it and beginner viewers cannot move into the expert one, whose continuation presupposes the whole background. The switch system acquires one-way transitions pointing from the realization that assumes more toward the

one that explains more, and they become two-way at the point where the explanation is complete. Time distance grows with every added explanation, because explaining takes time, and Chapter 9 considers where that time can come from.

7.3 Procedural transformation

A procedural transformation shows the same action at different granularity: a repair, a surgery, a calculation, carried out in full or with familiar steps compressed. It is a special case of exposition, distinguished because what varies is not background knowledge but the resolution at which an action is depicted. Its effects on switching are the same, and it has one of its own: coarse and fine realizations of the same action can share their first and last frames exactly, which makes the action's endpoints natural rendezvous.

7.4 Intensity transformation

An intensity transformation changes sensory or emotional load: how much violence is shown, how loud, how fast, how many flashes, how severe the suspense. It is often motivated by audience sensitivity rather than taste, and it is the transformation most often proposed for age-graded versions.

When done by cutting away or framing differently, it leaves fact distance near zero: the same event occurs, less of it is seen. When done by changing what happens, it becomes a causal transformation and inherits that kind's costs. Chapter 4 applies here with force. A lower-intensity realization does not protect anyone from the higher-intensity one on the same screen, because the composite contains both.

7.5 Perspective transformation

A perspective transformation follows different characters, or places the camera differently, through the same events. Its effect on fact distance depends on a single question: does one perspective show something the other does not? Two views of a conversation from opposite sides differ only in salience. A view that follows a character into a room the other view never enters differs in what it establishes. Perspective transformations can therefore be as cheap as tonal ones or as consequential as causal ones, and the difference is decided by information, not by camera position. Chapter 12 develops this.

7.6 Causal transformation

A causal transformation changes what happens: a character decides differently, an event occurs in one realization and not the other. It is the most powerful kind of divergence and the most expensive.

Fact distance grows with every consequence. Each realization establishes literals the other contradicts, and the realization graph thins accordingly. Unless the realizations reconverge, bringing their established facts back into agreement on everything later continuations depend on, switching is closed for good. Reconvergence is possible but has a narrative price: the story

must supply reasons why different decisions led to the same place. Stories built this way tend toward fatalism, in which choices change the route and not the destination, and a writer should recognize that the structure imposes the theme.

7.7 Terminal transformation

A terminal transformation is a causal transformation placed at the end: realizations that share everything until a final branch and then resolve differently. It avoids the price of reconvergence by never paying it. The realization graph is complete until the branch and empty after it, and image distance is zero for most of the running time, which makes terminal families nearly free in light.

The cheapness is partly deceptive. An ending changes the meaning of what came before, so a family with different endings is not really identical until the branch. The same scenes are seen differently depending on where they lead, and if the endings are to feel earned, the shared portion must contain the seeds of each. Chapter 10 distinguishes the ways of doing this.

7.8 Localization transformation

A localization transformation changes language, performers, cultural references, signage, or setting for different audiences. When it changes only language, it is close to tonal: same events, different surface. When it changes jokes, customs, or performers, it can alter what is established about characters and their world, and it acquires fact distance of its own. It also raises questions of authorship that the other transformations do not, because it asks one culture's realization to stand in for another's. Chapter 13 takes these up.

7.9 Composition and interference

Transformations are rarely used alone, and they interfere with each other. A family that varies both tone and explanation does not get four independent realizations for free. Comedy compresses and drama lingers, so the time available for explanation differs between the tonal variants, and an expert comic realization may leave no room for what a beginner dramatic realization needs. An ending that suits a comic realization may be unearned in a dramatic one. The nominal count of a family, the product of its dimensions, overstates what it actually contains whenever the dimensions interact.

The three distances are analytically independent, in the sense that each can be varied while the others are held fixed. They are not operationally independent, in the sense that real transformations rarely touch only one. Comedy alters timing as well as image. Explanation introduces facts as well as time. An alternate ending can demand different earlier imagery to seed it. If $\Delta d_i(T)$ is the change in distance i produced by a transformation T , then in general

$$\Delta d_i(T_a \circ T_b) \neq \Delta d_i(T_a) + \Delta d_i(T_b), \quad (7.1)$$

and variants that are cheap when designed separately can become expensive when combined.

Even so, the three distances give a way to anticipate the interference to first order. Tonal and localization changes mostly spend image distance. Expository and procedural changes mostly spend time distance, with one-way fact distance. Causal and terminal changes spend fact distance. Two transformations that spend the same distance compete for it; two that spend different distances can often be combined. A comic and dramatic family with beginner and expert realizations competes for time, and must be planned on the clock. A comic and dramatic family with two endings spends image distance throughout and fact distance only at the end, and combines easily, unless the endings need different seeds in the comic and dramatic realizations. The eight-way prototype of Appendix E, crossing genre, depth, and ending, exists precisely to measure these couplings.

7.10 A writer's summary

The grammar reduces to three questions a writer can ask of any proposed divergence. Will it show different pictures? Then it costs light. Will it establish different facts? Then it costs switching, until the facts are brought back into agreement. Will it take different amounts of time? Then it costs synchronization, until the time is repaid. Every difference in a variant family pays in at least one of these currencies, and the art of designing a family is largely the art of choosing which.

Chapter 8

Genre: The Rarely/Barely Pair

The idea behind this book began with a pair of films. *Rarely Needed Protocols* was to be a science-fiction thriller: a young astronaut named Theseus arrives on a planet in civil war, discovers that its ruling class controls the population through technology, sabotages the control systems, and must live with the chaos his sabotage unleashes. *Barely Needed Protocols* was to be a satirical comedy for children about the same world and the same events, mocking the absurdity of a regime that bans square televisions and permits only round ones. The two were to be screened at the same time, on the same screen, separated by the timing of the viewers' glasses, and joined by a soundtrack written to serve both.

This chapter takes the pair as the founding case of genre variation. It argues that genre is the most natural dimension for a variant family, and the one that most clearly shows why such a family has to be written together from the start.

8.1 Complementary, not restrictive

It is worth being precise about what the pair was and was not. It was not an adult film with a children's cut, in which material is removed so that a younger audience can safely see what remains. Chapter 4 showed that the glasses cannot enforce such a removal: a child without glasses sees the composite, and the adult material is in it. The pair was two complete tellings of one story, each addressed to its own audience, neither containing anything the other audience needed to be protected from.

The distinction is not only technical. A restrictive family is organized around what one audience must not see, and its variants are defined by absence. A complementary family is organized around what each audience is offered, and its variants are defined by address. The thriller treats the regime's control as a horror and Theseus's sabotage as a moral wager with real costs. The satire treats the same control as ridiculous and the same sabotage as a prank that goes gloriously wrong. Both are true to the events. Neither is a diminished form of the other. A parent and child who watch them side by side have seen the same story told to each of them.

8.2 Distances

In the vocabulary of Chapter 5, a genre pair has a distinctive profile.

Its fact distance is small by design. Theseus arrives, discovers the control systems, sabotages them, and faces the consequences in both realizations. What is established about the world and its events is the same, even when the reasons given for them differ in emphasis. The realization graph therefore stays close to complete, and a viewer can move between the thriller and the satire at almost any point without losing their place in the story. This is the strongest argument for genre as a first dimension: it gives the richest switching of any kind of variation.

Its image distance is large. Genre lives in performance, framing, lighting, and cutting, and all of these are on screen. Unless the pair shares footage, which is possible to a limited degree, the realizations show different pictures for most of the running time, and the costs of multiplexing in light and flash rate, examined in Chapter 3, are paid throughout.

Its time distance is where the work lies.

8.3 Genre keeps time

Comedy and drama do not only look different. They move differently. A joke depends on timing measured in fractions of a second: the pause before a reaction, the beat before a repetition, the interruption that arrives exactly too early. Drama often depends on the opposite, on holding a shot longer than is comfortable, letting silence do the work. The same event, played for laughs and played for weight, rarely takes the same number of seconds.

A variant family cannot require its realizations to match second by second, and it does not need to. It needs them to match at the points where it matters, and it can let them diverge in between. The natural unit is the scene. Assign each scene j an *envelope*, an interval $[s_j, e_j]$ on the common clock, and require every realization to begin and end the scene within it. Inside the envelope each realization keeps its own time. Write $\delta_k(t)$ for how far realization k is ahead of or behind the scaffold at time t . The envelope condition says

$$\delta_k(s_j) = \delta_k(e_j) = 0 \quad \text{for every scene } j, \quad (8.1)$$

while $\delta_k(t)$ may vary freely within the scene.

The envelope turns genre timing from an obstacle into a constraint that can be written to. A comic realization that needs three extra seconds for a reaction must give them back before the scene ends, perhaps by cutting a line the drama needs and the comedy does not. A dramatic realization that holds on a face must find the time in a transition the comedy rushes through. The envelope boundaries are also the natural candidates for structural rendezvous, because both realizations have completed the same action there, and the realization graph is most reliably complete at the moments the envelopes enforce.

8.4 Three ways to make a genre pair

There are three production strategies, and they trade image distance against cost and freedom.

The first uses the same footage for both realizations and varies only editing, music, and sound. This is the cheapest and the most constrained. Image distance is small, because many frames coincide, so the light cost is low and adaptive gating opens the glasses much of the time. But a comedy cut from dramatic footage, or the reverse, is limited by performances that

were given for one purpose. Editing can do a surprising amount, as trailers that recut a drama as a comedy demonstrate, but it cannot make a solemn actor look mischievous.

The second shoots the same blocking twice with different performances. The actors hit the same marks, the camera takes the same positions, the action happens in the same order, and the delivery differs: one take for the thriller, one for the satire. Image distance is moderate. Many shots coincide in composition and differ in expression. The strategy preserves the structural correspondence that keeps envelopes easy to meet, and it gives each realization its own performances.

The third stages the two realizations separately, each shot in its own way, joined only at the envelope boundaries. It gives the most freedom and costs the most, in production and in light, because the realizations share almost no frames. It is appropriate when the genres want genuinely different visual languages, which a thriller and a children's satire plausibly do.

Nothing requires one strategy throughout. A pair might share establishing shots and action, reshoot dialogue scenes with different performances, and stage a few set pieces separately. The choice can be made scene by scene, with the image budget in view.

8.5 Genre is written in

The strategies share a premise that is easy to miss. In each of them, the story must already be able to bear both readings. A thriller cannot be made into a comedy after the fact without a comedy's structure, and a satire for children cannot be turned into a thriller without a thriller's stakes. The events that both realizations share have to be events that are both funny and serious, or at least capable of being told as either. This is a constraint on the story before it is a constraint on the film.

It is also a real artistic opportunity. Stories that support both readings are not rare; much of the best comedy is about serious things, and much of the best drama has absurdity in it. The *Protocols* pair is an instance. A regime that controls its population by controlling the shape of its televisions is sinister and ridiculous at once, and each realization develops one side of a situation that already contains both. The strongest genre pairs will be those whose shared events contain the tension their realizations divide between them.

There is a characteristic difficulty. Comedy and drama sometimes disagree about what must be shown. A comic beat may depend on a visual detail the drama has no reason to include, and a revelation in the drama may need a close-up the comedy would find heavy. Where the difference is only in emphasis, both realizations can show the detail and weight it differently. Where one realization needs to establish something the other does not, the difference becomes fact distance, and the pair must either reconcile it at the next envelope boundary or accept that switching is closed until it does. The grammar of Chapter 7 predicts exactly this: genre variation is cheap in fact distance only as long as the genres agree on what happens.

8.6 The shared score

The original conception of the pair gave it one soundtrack, composed so that it would serve both films at once. That choice is examined fully in Chapter 11, but it bears on genre directly. Music is one of the strongest carriers of genre, and a score that serves a thriller and a satire simultaneously cannot simply be thrilling or funny. It must be ambiguous in the way the

shared events are ambiguous, supporting tension in one picture and absurdity in the other. That is difficult, and it is also a familiar achievement: film scores often carry irony, and a single cue can read as menace or mock-menace depending on what it accompanies. The shared score makes the genre pair a single work in the room even while the pictures diverge.

8.7 The composite of two genres

A final possibility follows from Chapter 4. A viewer without glasses, or with a coarser setting that admits both streams, sees the average of the thriller and the satire. Ordinarily that average is a double exposure of two performances. But a pair made with the second strategy, the same blocking with different delivery, produces a composite in which the same figures stand in the same places with two expressions superimposed. That composite is neither thriller nor comedy. It is something closer to a tragicomic ghost image, and it may be watchable in its own right as the room's shared view of an event that is both. Whether it is worth designing for is an artistic question. That it exists at all, as a third realization no one had to write, is a property only this medium has.

Chapter 9

Explanatory Depth

A second natural dimension of variation is how much a film explains. A documentary about a proof can pause to define its terms or assume them. An instructional film can show every step of a repair or compress the familiar ones. A historical drama can supply background or trust the audience to have it. Realizations that differ in this way are usually called beginner and expert versions, and this chapter uses those names while arguing that both are misleading.

9.1 What differs

Beginner and expert realizations do not differ in intelligence, and neither is a simplified form of the other. They differ in what they presume. The expert realization presumes that its viewers bring certain background knowledge; the beginner realization supplies that background on screen. Both can be equally demanding. The beginner realization asks its viewers to learn something during the film; the expert realization asks them to build on something learned before it.

In the terms of Chapter 6, the difference is in how each realization comes to hold its background. Call the background a realization needs B . The expert realization *declares* B : it treats it as established from the first frame. The beginner realization *establishes* B gradually, explanation by explanation, so that at time t it has established some part $B(t) \subseteq B$, growing until, at some time t^* , it has established all of it.

9.2 The one-way window

This structure produces the asymmetry noted in Chapter 7, and makes it exact.

Before t^* , a viewer of the expert realization holds all of B , by declaration, and the beginner continuation presupposes only $B(t)$. The switch from expert to beginner satisfies coverage. A viewer of the beginner realization holds only $B(t)$, and the expert continuation may presuppose any part of B . The switch from beginner to expert fails coverage whenever the expert continuation relies on background not yet explained. After t^* , the beginner viewer holds all of B , and the switch becomes possible in both directions, other things being equal.

The interval before t^* is therefore a *one-way window*. Transitions in the switch system run from the realization that assumes more toward the one that explains more, and t^* , the moment the explanation is complete, is where they become two-way. Writers of a depth family can

place t^* deliberately. A family that front-loads explanation reaches it early, and its realizations become mutually accessible for most of the film. A family that explains as it goes reaches it late, or never, and keeps the window open throughout.

The asymmetry has a practical meaning. A viewer who chose the expert realization and finds a passage too hard can move into the beginner realization during the window and follow the explanation there. A viewer who chose the beginner realization cannot jump ahead into the expert one until the explanation is done. Moving toward more help is always possible; moving toward less help must wait for the help to be complete. That is, on reflection, how learning works.

9.3 A presumption about people

The formal account has a weak point, and it should be named rather than hidden. Counting the expert realization's declared background as "established" is a presumption about its viewers. The film has not shown them B . It assumes they already have it. The switch system inherits the assumption: it treats any viewer on the expert realization as holding B , whether or not they do.

This matters in two ways. First, it means the switch system cannot protect a viewer who chose the expert realization without the background. That viewer will be lost, and the system will report that their history is coherent. Second, it means that whoever decides which viewers are on which realization is making a judgment about what those viewers know. If viewers choose for themselves, the judgment is theirs, and a wrong choice is a mistake they can correct by moving toward the beginner realization, which the one-way window always permits. If someone else assigns them, the judgment is made about them, and it can be wrong in ways that are not merely inconvenient. A student placed on the beginner realization by a teacher is being told something about themselves. Chapter 21 returns to this. The principle this chapter can state is that depth should be a property of the presentation a viewer selects, not a rank assigned to the viewer.

9.4 Where the time comes from

Explanation takes time. A beginner realization that pauses to define a term, show an intermediate step, or supply background runs longer than an expert realization that does not, unless the time is found somewhere. In the vocabulary of Chapter 5, explanation spends time distance, and the envelopes of Chapter 8 require it to be repaid before each scene ends. There are four places to find it.

The first is redundancy. Films contain time that carries little information: establishing shots, transitions, lingering reaction shots. A beginner realization can replace some of it with explanation. The cost falls on atmosphere, which the beginner realization may miss.

The second is to use the same minutes differently. An expert realization need not be a beginner realization with the explanations cut out, a shorter film padded to length. It can spend the time the beginner realization uses for explanation on something the expert viewer will value: a further implication, a harder example, a historical aside, a complication the beginner realization cannot afford. The two realizations then have equal length and different content, and each is complete on its own terms. This is the strongest arrangement, because it makes the

expert realization a different film rather than a lesser one, and it keeps the two synchronized without either waiting for the other.

The third is sound. Much explanation is verbal, and verbal explanation can in principle be carried by a separate audio channel while the picture stays common. This requires individual sound, and Chapter 11 shows that individual sound has costs of its own. If the family keeps a shared score, explanation must be carried by the picture: diagrams, captions, demonstrations, visual emphasis. That is a real constraint, and it shapes what a depth family can be. A shared-sound depth family is closer to a well-designed diagram than to a lecture.

The fourth is flexibility at the boundaries. A family can leave some intervals before its rendezvous deliberately loose, with material in the expert realization that can be shortened or lengthened to absorb whatever time the beginner realization needs. This is the least elegant solution and often the most practical.

9.5 Presentation is not uptake

There is a deeper limit, and it concerns what “established” means. The model of Chapter 6 counts a proposition as established when the film presents it as settled. But presentation is not uptake. A fact shown in the corner of a frame, or mentioned once in a line of dialogue, is presented to every viewer and taken up by some. Depth variation is often less about what is presented than about whether it is taken up: the beginner realization shows the same thing more slowly, more prominently, more than once.

Mystery films make the point sharply. A beginner realization of a mystery might foreground its clues, holding on them, returning to them, so that a viewer is likely to notice. An expert realization might present the same clues in passing, leaving the viewer to catch them. In the literal model the two realizations establish the same facts, and the realization graph is complete. In practice they differ. A viewer who watched the expert realization throughout and missed a clue is, by the model, fully prepared for its solution, and may be baffled by it. A viewer who switches into the expert realization from the beginner one at the solution arrives better prepared than the viewers who were there all along.

The model can be extended to track salience, weighting what is established by how prominently it was shown, but the extension is uncertain, because uptake varies between people. It is better to state the limit plainly. Switch admissibility guarantees that a viewer was shown what they need. It cannot guarantee that they took it in. For most variation this gap is small. For depth variation it is the whole subject, and the switch system should be read as a floor on coherence, not a promise of comprehension.

9.6 Depth as a gift

The chapter’s argument can be summarized by what depth variation offers when it is done well. Every viewer in the room follows the same film, in step, at the level they chose. Those who want more explanation receive it without slowing anyone else, and those who want less receive something in its place rather than a pause. Any viewer who finds their choice too hard can move toward more help at any moment, and anyone who chose more help can move toward less once the help is complete. Nobody is ranked. That is a considerable thing to be

able to offer in a single screening, and it is available only because the explanation happens in time a common screen would otherwise have to share.

Chapter 10

Endings and Retrospective Meaning

Alternate endings are the most familiar kind of variation. Films have been released with different endings for different markets, recut after test screenings, and restored in director's versions with their original conclusions. *Clue* (1985) went further, distributing three endings so that different theaters showed different ones, and which ending a viewer saw depended on where they went [2]. That was the intertheater variation of Chapter 13, applied to endings. Selective cinema changes one thing about them: the endings are seen at the same time, in the same room, by people who then leave together. This chapter examines what an ending does to the film before it, and what a family of endings does to an audience.

10.1 The branch

The simplest family of endings shares everything until a final branch and then resolves differently. Before the branch, the realizations are identical, or differ only in ways that establish nothing relevant. After it, they establish contradictory things: the character survives in one and dies in the other, the plan succeeds or fails, the suspect is guilty or innocent.

In the terms of Chapter 6, the branch is the family's last structural rendezvous. Before it, the realization graph is complete. After it, the graph is empty: a viewer who has seen one ending begin cannot move into another without being told something that contradicts what they have just seen. The branch point itself is a moment of choice. A viewer who turns the dial at the branch, and only there, selects an ending without interrupting the film and without any incoherence. Selective cinema therefore provides something branching video provides only by stopping the story: a choice of ending made in the flow of the screening, by each viewer independently, with nobody else in the room affected.

In image distance the branch family is the cheapest of all. Its realizations coincide until the branch, so adaptive gating keeps the glasses open for most of the film, and the costs of multiplexing are paid only in the last minutes. It is the natural first project for anyone building such a system.

10.2 What endings can be

An ending cannot be anything at all. It must be consistent with what the film has already established.

Proposition 10.1. Let a family of endings share everything before a branch at time t_b , and let $E_{\text{sh}} = \text{Est}(F^{\leq t_b}) \cap \mathcal{R}$ be what the shared portion establishes that any ending depends on. Every coherent ending commits only to literals consistent with E_{sh} .

Proof. Each realization must admit its own continuation. An ending committing to the negation of a literal in E_{sh} would fail consistency against its own history. $\square$

The proposition is simple and its consequence is not. The space of possible endings is bounded by what the shared portion leaves *unestablished*. Everything the film settles before the branch is common to every ending; everything it leaves open is where the endings can differ. A writer's control over a family of endings is therefore, to a large extent, control over what the film declines to settle. A shared portion that establishes too much leaves the endings no room. One that establishes too little leaves the story shapeless. The craft is in presenting a great deal while settling only what every ending can share.

10.3 Terminal, seeded, and retrospective

There are three ways to use that space, and they distribute fact distance differently.

A *terminal* family places all of its divergence after the branch. The shared portion is neutral with respect to the endings; nothing before the branch points toward one rather than another. The fact distance is concentrated in the final minutes, which keeps switching open until then. The risk is that the endings feel arbitrary. An ending the film has not prepared for can feel attached rather than arrived at.

A *seeded* family prepares each ending before the branch. Seeds can be placed in two ways. If each realization carries only the seeds of its own ending, the realizations diverge before the branch, establishing different foreshadowing, and the last structural rendezvous moves earlier. Fact distance is distributed backward through the film, and switching closes sooner. If instead the shared portion carries the seeds of every ending, the realizations stay identical until the branch, but the shared portion must support several futures at once. It becomes deliberately underdetermined, rich in possibilities none of which it confirms. Seeding in this second way is closer to writing a good mystery than to writing a single story, and it is the harder and more interesting form.

A *retrospective* family shares its footage almost entirely and lets the ending change what the footage meant. A final disclosure recasts earlier scenes: a helpful character turns out to have been manipulating everyone, or a threat turns out to have been a misunderstanding. The same shots are seen again in memory, differently. This works only because of the proposition above. The earlier scenes presented the character's helpfulness without establishing it as fact; they established that the character acted helpfully, and left the motive unsettled. Every ending is consistent with what was shown because what was shown was carefully less than what was implied. Retrospective endings live in the gap between presentation and establishment that Chapter 9 identified from the other side.

10.4 An ending is not a scene

It follows that two realizations with different endings are not identical until the branch, even when their frames are. The same shot of a character's face means one thing in a film that

ends in betrayal and another in a film that ends in loyalty. Viewers will not know which they are watching until the end, but afterward they will remember the earlier scenes through the ending they saw. Fact distance, as defined so far, is a property of what the film establishes as it goes. Meaning also has a retrospective component that the definition does not capture: the ending reaches back and changes the significance of everything before it, without changing a single literal the earlier scenes established.

This is why a family of endings is more than a film with a choice at the end. Viewers of different endings have, in a real sense, seen different films throughout, even though they looked at the same frames for most of the running time. The divergence was postponed, not avoided.

10.5 Leaving together

The social consequence is unusual. Two people who watched different endings leave the theater having shared nearly everything and resolved it differently. Their conversation does not begin with what they thought of the ending, as it would after an ordinary film. It begins with which ending each of them saw.

For viewers who did not know the endings differed, that discovery can be disorienting, even unwelcome: the realization that a companion was shown something else. For viewers who knew, it can be the best part of the work. They compare, reconstruct each other's endings, and reinterpret the shared portion in light of both. The film is completed in the conversation. Chapter 18 takes this up and argues that the strongest uses of the form may treat the post-screening conversation as its final act.

The ending is also where the form is most fragile. A leak of a fraction of a second from another stream, which Chapter 2 showed is hard to eliminate entirely, costs almost nothing in the middle of a genre pair. In the last minute of a family of endings, a glimpse of a face that should be dead can ruin the conclusion. The acceptable level of leakage is not uniform through the film. It should tighten as the ending approaches, and the blanking intervals, light budget, and exhibition settings may need to change with it.

10.6 The ending nobody chose

Chapter 4 showed that the room without glasses sees the average of the streams. In a family of endings, that composite is quiet for most of the film, because the streams coincide and the unaided viewer sees the film correctly. At the branch it becomes something else: every ending superimposed, the survivor and the corpse at once, the plan succeeding and failing in the same frame.

Ordinarily this would be a defect to minimize. It could also be designed. The composite of all endings is the undecided ending, the room's own version of the conclusion, in which nothing has yet been settled. A viewer who lifts the glasses at the branch sees the story before it chooses. Composing endings so that their average is itself meaningful is difficult, subject to the one-stream cost of the proposition in Chapter 4, and possible only where the endings share their staging. But where it can be done, it gives the form an image no other medium has: a single frame containing every way the story could have ended, visible to anyone who looks without choosing.

Part III

Sound

Chapter 11

The Composite Soundtrack

Everything so far has concerned light. Light can be divided in time, and glasses can select the parts. Sound cannot be divided that way. A loudspeaker fills the room, every ear in the room receives what it emits, and no shutter can admit one second of it and reject the next without making nonsense of both. A selective screening can show different people different pictures. Unless something more is done, it plays everyone the same sound.

This chapter treats sound as the harder multiplex. It sets out the four ways a variant family can handle sound and defends the one that was part of the idea from its beginning: a single score composed to serve every realization at once.

11.1 Four arrangements

Common sound plays one soundtrack in the room for everyone. It keeps the room acoustically whole and constrains the pictures severely, because every realization must fit the same sound.

Individual sound gives each viewer the soundtrack of their own realization through headphones, earpieces built into the glasses, or bone-conduction transducers. It allows the realizations to differ completely, and it turns the room into a set of private listening booths.

Hybrid sound divides the soundtrack. Music and ambient effects play in the room; dialogue, or narration, or commentary, reaches each viewer individually. It keeps some of the room's shared sound and releases the realizations from sharing their words.

Complementary sound plays one soundtrack in the room, like common sound, but composes it for the family rather than for any one realization. It is written to be right for every picture, meaning something different under each.

The first three are engineering choices. The fourth is an artistic one, and it is the one the original *Protocols* pair proposed: two films watched at the same time with “overlapping, complementary soundtracks.”

11.2 What individual sound costs

Individual sound is technically easy. Several audio channels can be broadcast to receivers in the glasses as readily as the timing signal, and the glasses remain devices that only output, recording nothing. The costs are not technical.

The first is isolation. A room in which everyone wears headphones is not quite a cinema. It is a set of people each hearing a private film, and much of what the Introduction called the shared occasion is gone. The picture is common, or at least commonly placed, but the sound, which carries more of a film's emotional timing than viewers usually notice, is not.

The second is subtler, and it applies to every arrangement: the audience is also a sound source. People laugh, gasp, go silent, shift in their seats. In an ordinary screening these sounds are part of the experience, and they are synchronized with it, because everyone is reacting to the same thing. In a selective screening they are not. The viewers of the comedy laugh at a moment when the viewers of the drama are watching a death. Headphones do not prevent this, because the laughter is in the room. The audience's own reactions are a shared channel that the film does not control.

The third cost is to the unfiltered view. A viewer without glasses sees the composite of the pictures. With individual sound, that viewer hears nothing of any realization's dialogue, or hears one realization's sound over everyone's pictures. The room's own version of the film, which Chapter 4 argued could be designed, loses its soundtrack.

11.3 The dialogue constraint

Common and complementary sound both require the realizations to share whatever is audible, and the hardest thing to share is speech. Music can be ambiguous. Effects can serve several pictures. But words are words, and delivery is sound: the tone of a line is in the audio, not in the image.

This produces a precise constraint. If a face is seen speaking in any realization, the shared soundtrack must carry speech that matches its lips. So wherever two realizations both show a speaking face at the same moment, they must show it saying the same words, with the same timing. They can differ in what surrounds it, in framing, in reaction, in what the listener's face does, but not in the speech itself. Under shared sound, divergent dialogue is possible only where mouths are not visible: speech off screen, from behind, in wide shots, or through voice-over.

The constraint has a direct consequence for Chapter 8. The second production strategy for a genre pair, the same blocking shot twice with different performances, relies on different delivery. Under shared sound, the delivery cannot differ. The pair must either rely on visual performance, expression and gesture and timing of reaction, while the lines themselves are spoken once, or move divergent dialogue off screen, or adopt hybrid sound. None of these is fatal, and the first is the most interesting. A line written so that one delivery can be heard as menace in one picture and as bluster in another is a small version of the complementary score, and good actors can give such readings.

11.4 Complementary scoring

The complementary score rests on a familiar fact about film music: its meaning depends on the picture. A slow, dissonant cue under a character climbing stairs in the dark is dread. The same cue under a character creeping to raid a refrigerator is comedy. Composers exploit this constantly, and audiences read music through what they see without noticing that they do.

Studies of film sound have described the relation in both directions: music shapes how an image is read, often without being consciously heard [6], and sound adds meaning to an image that viewers then attribute to the image itself, what Chion calls added value [1]. A complementary score relies on the same relation run once against several images.

A complementary score makes the exploitation systematic. It is written against several pictures at once, so that each cue is right under each realization. Formally, if A_k is the set of soundtracks acceptable for realization k , the shared score must lie in their intersection,

$$S \in \bigcap_{k=1}^m A_k, \quad (11.1)$$

and the art is in finding a score that is not merely acceptable under each picture but good under each. The intersection is small when the realizations want incompatible things from sound, loud where the other wants silence, fast where the other wants stillness, and it is larger than it looks when the realizations share the events that sound marks. The *Protocols* pair is favorable: a thriller and a satire about the same regime want tension from the same moments, and a score of ominous, slightly overstated cues can be dread under one and parody under the other.

The complementary score does something neither individual nor hybrid sound can. It keeps the room acoustically single, so that everyone hears the same music at the same moment, and it makes that single sound mean different things to different people. It is the audio counterpart of the designed composite of Chapter 4, with one difference: the composite image is what the room sees only without glasses, while the complementary score is what everyone hears all the time. It is the most thoroughly shared element of a selective screening, and it is where the family is most visibly one work.

11.5 Scheduling the audience

The complementary score also suggests a remedy for the audience's own sounds. If the realizations are composed together, their emotional peaks can be placed together. A comic realization can put its laughs where the dramatic realization permits release, and a dramatic realization can place its silences where the comedy is between jokes. The envelopes of Chapter 8 give the structure for this: moments of shared tension and shared relief can be aligned at their boundaries, so that the room's laughter falls where nobody is being asked to grieve.

This cannot be made perfect, and it should not be. Some dissonance between what a viewer sees and what they hear the room doing is part of the experience of selective cinema. A viewer of the drama who hears laughter from part of the room at a solemn moment learns, audibly, that others are watching something else. That knowledge is a large part of what Chapter 17 will call synchronized difference. The aim of scheduling is not to hide the difference but to keep it from becoming cruelty.

11.6 Choosing an arrangement

The four arrangements trade divergence for community. Individual sound permits the most divergence and keeps the least of the room. Common sound keeps the room and permits the

least. Hybrid sound sits between them. Complementary sound keeps the whole room and permits more divergence than common sound, at the price of composing for the family from the start.

A family's sound arrangement should therefore follow from what kind of family it is. A tonal family, whose realizations share events, suits complementary sound well. A depth family, whose extra explanation is often verbal, pushes toward hybrid sound, or toward carrying explanation in the picture, as Chapter 9 noted. A family of endings can share its score until the branch and needs a composer who can end in several directions at once. A family of unrelated films on one screen, the weakest case in Chapter 5, needs individual sound and gives up most of what makes the form worth building.

The early notes behind this book did not argue for complementary sound. They simply assumed it: the two films would be watched simultaneously "as they have overlapping, complementary soundtracks." The assumption turns out to carry the idea's social promise. A selective screening with shared, complementary sound is one event that people experience differently. A selective screening with private sound is several events that happen to share a room.

Part IV

Space and Place

Chapter 12

Selectable Viewpoint

A film is shot from somewhere. The camera's position, its movement, and the editing that joins its shots decide what the audience sees of events that, in the fiction, happen all around. A variant family can vary that decision: one realization follows one character, another follows a second; one watches a scene from the doorway, another from across the table. This chapter asks when such variation is only a matter of angle and when it becomes a different story.

12.1 Scenes and cameras

It helps to separate a scene from the ways of filming it. Describe a scene by a model

$$\mathcal{S} = \langle G, E, C, T \rangle, \quad (12.1)$$

where G is the geometry of the place, E is the set of events that occur, C is the set of available camera positions and paths, and T is the scene's own timeline. A realization of the scene selects, for each moment, a camera path $c_k(t) \in C$ and an editorial policy for cutting between paths. Two realizations of one scene model share its geometry, events, and timeline and differ only in how they are seen.

Selectable viewpoint in a cinema is modest compared with the freedom of virtual reality, where every viewer steers their own camera. The screen is shared and the number of streams is small, as Chapter 3 showed, so each stream is one authored camera path, and the family offers a few of them. What it offers instead is authorship: each viewpoint is composed and edited, not wandered.

12.2 Angle and information

The key distinction is between viewpoints that change *salience* and viewpoints that change *information*.

Two cameras on opposite sides of a conversation show the same events. One favors the speaker's face and the other the listener's, and viewers of each will feel the scene differently, attending to different reactions. But nothing is established in one that is not established in the other. Such viewpoints differ in image distance and leave fact distance at zero.

A camera that follows a character out of the room, while the other realization stays behind, is different. It shows events the other does not: what the character did in the corridor, who they

met, what they overheard. Now the realizations establish different things. Let $\text{vis}(c, e)$ indicate whether event e is visible, or audible, from camera path c . What a viewpoint establishes is essentially what it lets the viewer witness, and two viewpoints differ in fact distance exactly when they differ on events that some continuation depends on:

$$\{e \in E : \text{vis}(c_a, e)\} \cap \mathcal{R} \neq \{e \in E : \text{vis}(c_b, e)\} \cap \mathcal{R}. \quad (12.2)$$

So camera position is not, in itself, what matters. A radically different angle can leave the story untouched, and a small change, a door left open or closed, can change what a viewer knows. Selectable viewpoint is cheap narratively when it varies salience and as consequential as a causal branch when it varies information. The grammar of Chapter 7 placed perspective transformation between the tonal and the causal for this reason.

12.3 Point-of-view families

The most interesting viewpoint families follow different characters. Each realization is the story as one character lives it: what they see, where they go, whom they meet. Such families have a natural structure that the switch system makes visible.

While the characters are apart, their realizations witness different events and establish different things. Fact distance grows, and switching closes. When the characters meet, in the same room, witnessing the same events, their realizations witness the same things and fact distance stops growing. Whether it falls back to zero depends on whether what each character learned apart has been shared; a conversation in which they tell each other what happened does exactly that. So in a point-of-view family, structural rendezvous tend to fall at scenes where the point-of-view characters are together and have caught each other up. The story's meetings are the film's switching points.

This gives writers a clear handle. A point-of-view family that wants viewers to be able to move between characters must bring its characters together, and must let them exchange what they have seen. One that wants each realization to stay private, so that viewers of each character's story keep secrets from each other, can keep the characters apart and let the switch system stay closed. Both are legitimate. They are different films.

12.4 Salience without information

Viewpoints that vary only salience deserve more attention than their narrative cheapness suggests. They are the purest case of what selective cinema offers: the same events, the same facts, the same timing, seen differently. A family of salience viewpoints keeps the realization graph complete, so viewers can move freely between them. It keeps time distance at zero, because the events take the same time however they are seen. It spends only image distance, and that is exactly the budget adaptive gating recovers whenever the viewpoints happen to coincide.

What such a family changes is attention and sympathy. A viewer who watches a quarrel from one party's side and a companion who watches it from the other's will leave having seen the same argument and taken different sides. Nothing either was told contradicts what the other was told. Their disagreement is about the meaning of shared facts, which is the ordinary condition of people who were in the same room, and a salience family reproduces it inside a film. Cinema has long dramatized the related case in which accounts of one event conflict,

most famously in *Rashomon* [13]. A salience family is the gentler and in some ways stranger case, in which the accounts do not conflict at all and still produce different verdicts.

12.5 Capture and rendering

Selectable viewpoint can be produced in several ways. Multiple cameras on set can record the same take from several positions, which fixes the performances and lets the realizations share them. Volumetric capture records a scene as geometry and texture, from which camera paths can be chosen later. Virtual production and synthetic environments allow any camera path through a rendered scene.

The more flexible methods make viewpoint cheap to vary after the fact, and they raise a question that returns in Chapter 25: if camera paths can be chosen after shooting, the work is no longer a fixed set of shots but a scene model with rules for viewing it. Selective cinema does not require this, and most of what this chapter describes can be done with ordinary multi-camera shooting. But it sits naturally alongside it, because a scene model is exactly the object from which a family of viewpoints is drawn.

12.6 Two angles at once

The composite of two viewpoints, seen without glasses, is usually the least meaningful of all composites: two angles on a room superimposed, walls and faces at incompatible positions. It is rarely worth designing. But the failure is instructive. The composite of a genre pair or a family of endings could be designed because the realizations shared their staging. Viewpoint families by definition do not. They share events and differ in geometry, and averaging does not respect geometry. The one exception is small differences of angle, a camera shifted a little to one side, whose composite reads as depth rather than confusion. That exception is, of course, stereoscopy, and it returns the apparatus to where Chapter 2 found its ancestor.

Chapter 13

Local Realization

The variation considered so far happens inside one room: several viewers, several realizations, one screen. There is a second kind of variation that selective cinema makes newly possible, between rooms rather than within them. A work distributed as a scene model rather than a fixed sequence can be realized differently in different places, fitted to the theater, the language, the culture, even the performers. This chapter examines local realization and the ways it combines with selection inside the room.

13.1 Two axes of variation

Distinguish *intertheater* variation, in which different venues show different realizations, from *intratheater* variation, in which one venue shows several realizations to different viewers. Ordinary distribution already has the first in a crude form: dubbed and subtitled prints, regional cuts, censored versions. Selective cinema adds the second. The two can be combined, so that each venue offers its own local family, and a work becomes a family of families.

13.2 Fitting the room

The mildest form of local realization adapts the work to the physical theater. Screen size and shape, seating distance, and the height of the rows affect how an image reads. A close-up that dominates a small screen overwhelms a very large one; fine detail visible from the front rows is lost at the back. A work distributed as a scene model with declared camera rules can be rendered for each room: framing adjusted to the aspect ratio, camera distance to the typical viewing distance, pacing of dense visual information to the size of the screen.

This is variation that changes nothing a viewer is told. It adjusts image distance between theaters and leaves fact distance at zero, and it is uncontroversial except for one thing: it means that no two venues show quite the same frames. The work that critics review, archives preserve, and courts cite is no longer a single sequence. Chapter 25 takes up what the work then is.

13.3 Language

The most common local variation is language, and it collides immediately with Chapter 11. Language is carried by sound. A multilingual room watching one picture needs either individual sound, one language per viewer, or a shared track in one language with the rest subtitled. Subtitles are a visual variation, and a selective screening can carry them as separate streams: the same picture with different subtitles for each language, selected by the glasses. This is cheap in image distance, because only the text differs, and adaptive gating keeps the glasses open whenever no subtitle is on screen.

Dubbing is harder. A dubbed line does not match the lips that were filmed speaking the original. Under shared picture and individual sound, the mismatch is the ordinary one of dubbed film. Under individual pictures, it could be removed: each language stream could show lips synchronized to its own language, re-rendered or re-performed. This is now technically possible with synthetic re-rendering of faces, and it raises the question of performer consent, taken up in Chapter 22: an actor who spoke one line has not thereby agreed to be shown speaking another.

13.4 Local performers

A stronger form of local realization replaces the performers. A story's roles, blocking, and timing are held fixed, and each region casts its own actors, who perform the same scenes in their own language, with their own gestures, in settings adjusted to local architecture and custom. The result is more than dubbing, because the visible performers are local, and less than a remake, because the structure is shared and the local realizations remain synchronized with a common production.

This raises a question about performance. If several actors play one character in one structural role, the character becomes partly separable from any single embodiment. That separation is familiar from theater, where many actors play Hamlet, and unfamiliar in cinema, where a role is usually fixed to the face that played it. Local realization moves cinema toward theater in this respect. It also cuts the other way: a local performer may change the character substantially, and to call the result a "substitution" can understate what the performer contributed. Credit, in such a system, has to follow the realization, not only the structure.

13.5 Translation, localization, rewriting

Local variation comes in degrees, and the degrees differ in what they claim.

Translation renders the same meaning in another language. It aims at zero fact distance.

Localization replaces references that would not be understood: a joke about one country's politics with a joke about another's, a sign in one script with a sign in another, a food, a currency, a holiday. It aims at the same effect by different means, and it may change what is established, in small ways, to keep the effect.

Substitution replaces performers and settings while keeping the structure.

Rewriting changes the structure itself to fit the local culture: different relationships, different motives, different outcomes.

The further along this sequence a local realization goes, the less plausible it is to call one version the original and the others adaptations. A rewritten local realization is a different work that shares a skeleton with the others. Selective cinema does not create this problem; it has always existed in the relation between a film and its remakes. It makes the problem simultaneous, because the realizations are designed and released together, and it makes it visible, because a single venue may offer several.

13.6 The template problem

Local realization carries a risk that needs to be named. A work designed as a common structure with local surfaces presupposes that the structure is neutral: that its story, its relationships, its sense of what is funny or moving or just, can be carried into any culture by changing the faces and the signs. That presupposition is rarely true. A structure is written somewhere, by someone, and it carries the assumptions of its origin. Local realizations built on it import those assumptions while dressing them in local detail, and the local detail can make the import harder to notice.

The danger is a kind of cultural template: one culture's story with everyone else's faces as skins. It is not an argument against local realization. It is an argument that local realization should reach, where needed, as far as rewriting, and that the people who realize a work locally should have authority over its structure as well as its surface. A system that lets local creators change the faces but not the story has decided in advance whose story it is.

13.7 Local families

The two axes combine. A venue can offer, in one room, a local family: realizations in two languages, or a comic and dramatic pair performed by local actors, or a beginner and expert realization adapted to a local curriculum. Everything in Parts I and II applies within each venue, and local realization applies between them.

The combination also sharpens a point that runs through the book. Selective cinema makes it possible for people in different places, and people in the same room, to be shown different things under the same title. When the differences are declared, chosen, and made by people with standing to make them, this is an extraordinary enlargement of what a film can do. When they are not, it is a means of telling different audiences different things while each believes it saw the work. Part VIII is about the difference.

Part V

Authoring

Chapter 14

The Variant Matrix

A conventional screenplay describes one film. It lists scenes in order and, within each, what is seen and said. A variant family cannot be written this way, because what matters most about it is not visible in any single realization: what the realizations share, where they differ, what each establishes, and where they come back together. This chapter proposes a writing format that makes those relations explicit, the *variant matrix*.

14.1 Spine and columns

The matrix has two parts. The *spine* is the sequence of what all realizations share: the scenes, their envelopes on the common clock, the events every realization depicts, and the structural rendezvous. It is written first, and it is the family's skeleton. The *columns* run beside the spine, one per realization, and record how each realization renders each scene: its performance, framing, dialogue, explanation, and whatever else varies.

A reader of the matrix reads down the spine to follow the story everyone sees, and across a row to see how one moment is realized in each version. Neither reading alone is the work. The spine without the columns is an outline; a column without the spine is one film with its dependencies hidden.

14.2 What each row records

Each row of the matrix is a production unit, usually a scene, and records four things beyond its content.

Write the unit as

$$U_i = \langle A_i, V_i, \Delta_i, Q_i \rangle. \quad (14.1)$$

A_i are the unit's *anchors*: its envelope on the common clock and any moments within it that all realizations must meet, such as a shared sound event or the start of a common shot. V_i are the *permitted variants*: how each realization is allowed to render the unit, and which transformations of Chapter 7 it uses. Δ_i is the *state difference* the unit leaves behind: for each realization, the literals it establishes and the presuppositions it relies on, in the terms of Chapter 6. Q_i are the unit's *reconciliation requirements*: which differences must be brought back into agreement, and by which later unit.

From these four fields, most of what earlier chapters described can be computed rather than guessed. The switch system follows from the established and presupposed literals. The structural rendezvous are the units after which every realization's established set agrees on $\mathcal{R}$. The time budget follows from the envelopes and the realizations' planned durations within them. The light budget follows from which shots coincide across columns.

14.3 Checks

The matrix supports a set of checks a writer can run on a draft, by hand or with software.

A *coverage check* asks, for every unit and realization, whether what the unit presupposes has been established earlier in that realization. It catches a realization referring to something it never showed.

A *consistency check* asks whether any realization commits later to the negation of something it established. It catches a realization contradicting itself.

A *switching check* computes the switch system and shows where viewers can and cannot move. It catches families that were meant to be freely switchable and have quietly closed, and it shows where one-way windows open and close.

An *envelope check* sums each realization's planned durations within each envelope and flags any that do not fit.

A *dialogue check* flags every moment at which a face is seen speaking in more than one realization, and, if the family uses shared sound, confirms that the words match, as Chapter 11 requires.

A *seed check*, for families of endings, confirms that each ending's preparation is present in the realizations that lead to it and absent where it would give the ending away.

A *contamination check* flags information that one realization must keep from another's viewers, and asks where that information could escape: in a shared shot, in the composite of Chapter 4, or in leakage through the glasses near a transition. A clue meant only for the expert realization is contaminating if it appears in the background of a shot both realizations share.

None of these checks settles whether the family is good. They settle whether it is coherent, which is a precondition that is very hard to meet by intuition once a family has more than two realizations.

14.4 Writing order

The matrix suggests a writing order. The spine comes first: the events all realizations share, placed on the clock, with the rendezvous marked. The columns come next, written against the spine and against each other. Then the checks, and then revision, which will often move material between the spine and the columns. A scene that the realizations were meant to render differently may turn out to need a shared core to keep a rendezvous possible. A scene meant to be shared may need to split because one realization's genre or depth cannot live with it.

It follows that the family's writers must work together, not in sequence. A column cannot be written by a separate team after the spine is fixed, because the column's choices constrain the other columns through the envelopes, the dialogue constraint, and the switch system. The family is written as a whole or it does not cohere.

14.5 The matrix as a record

The matrix is also the family's most important record. It states what the family is: which realizations exist, what each is permitted to be, how they correspond, and where they agree. Chapter 25 argues that the identity of a selective work lies in exactly these declared constraints rather than in any single sequence of frames. If that argument holds, the matrix is not a production document to be discarded after release. It is the closest thing the work has to a text, and it should be preserved with at least the care given to the footage. Appendix B sketches a notation for it.

Chapter 15

Production Architecture and Coordination Cost

A variant family must be made, and making several related films costs more than making one. How much more depends less on the number of realizations than on how they are produced and how much work it takes to keep them consistent with each other. This chapter compares production strategies and argues that the dominant cost of a variant family is coordination, not footage.

15.1 Four strategies

Separate productions make each realization as its own film, sharing only the script's spine. They give the most freedom and cost the most, roughly as many films as there are realizations. They also share the fewest frames, so they are the most expensive to exhibit, in the light budget of Chapter 3.

Shared master photography shoots one set of footage covering the events common to all realizations and adds variant inserts: alternate performances of key moments, reaction shots, additional explanation, different endings. Most frames are shared, which keeps exhibition cheap and adaptive gating open. Freedom is limited by what the master footage allows.

Modular production breaks each scene into parts that are shot once and parts that vary. Invariant action, establishing shots, and anything whose meaning does not change are shot once; dialogue, reactions, explanation, and anything carrying genre or depth are shot in variants. It is shared master photography applied at the level of the shot rather than the scene, and it maps directly onto the spine and columns of the variant matrix.

Virtual and generative production builds scenes as models from which realizations are rendered: camera paths, lighting, sometimes performances. It makes variation cheap after the fact, especially variation of viewpoint and setting, and it moves cost from shooting to building and maintaining the models. It also creates new dependencies on software and asset preservation that the other strategies do not have.

A real production will mix them: modular shooting for most scenes, separate staging for a few set pieces, rendering for backgrounds or viewpoints.

15.2 The coordination term

A rough cost model for a family $\mathcal{F}$ of m realizations is

$$C(\mathcal{F}) = C_0 + \sum_{k=1}^m C_k + \sum_{a < b} C_{ab}, \quad (15.1)$$

where C_0 is the cost of what all realizations share, C_k is the additional cost of realization k alone, and C_{ab} is the cost of keeping realizations a and b consistent with each other: checking their envelopes, their dialogue under shared sound, their established facts, their composite, their switching.

The first two terms are the familiar costs of making films. The third is new, and it grows with the number of pairs, which is $m(m-1)/2$: one pair for two realizations, three for three, six for four, forty-five for ten. Even when the footage for an extra realization is cheap, the obligations it creates with every other realization are not. This is the main reason, beyond the optical limits of Part I, that variant families should stay small.

15.3 Coordinating through the spine

The quadratic growth is not inevitable everywhere, and the variant matrix shows where it can be avoided.

At a structural rendezvous, the requirement is that every realization agree on $\mathcal{R}$. Agreement is transitive: if each realization agrees with the spine, they agree with each other. So at rendezvous, each realization can be checked against the spine alone, and the coordination cost there is linear in m rather than quadratic. The same holds for envelopes, which are fixed on the spine's clock, and for shared-sound events, which are fixed in the shared track: each realization is checked against the fixed reference, not against every other realization.

Between rendezvous, where realizations diverge, the checks that matter are genuinely relational. Whether a viewer can move from one realization to another depends on what both have established, and the switch system of Chapter 6 must be computed over pairs and, strictly, over histories. The designed composite of Chapter 4 couples all realizations at once. These costs cannot be routed through the spine.

The practical lesson is structural. A family whose realizations meet the spine often, at frequent rendezvous with fixed envelopes and shared sound events, keeps most of its coordination linear. A family whose realizations diverge for long stretches pays the quadratic cost for the whole of each divergence. Frequent rendezvous are not only good for switching. They are how a family stays affordable.

15.4 Variant inflation

The coordination term also explains why the number of realizations a family advertises is a poor measure of what it offers. A family that crosses three independent dimensions, say two genres, two depths, and two endings, has a nominal count of eight. If the dimensions really were independent, its coordination cost would reflect eight realizations and twenty-eight pairs. In practice, as Chapter 7 showed, the dimensions couple, and some of the eight combinations will be close to each other while others are expensive to keep coherent.

There is a converse temptation: to inflate the nominal count with variations that cost little because they differ little. Four subtitle languages crossed with two endings is “eight versions,” and the eight differ from each other hardly at all. Describing a family by its count invites both errors. It should be described by its dimensions and, for each, by the distances it spends. Chapter 22 returns to this as a matter of honest description.

15.5 The family’s crew

Coordination needs people. A variant family needs a role that ordinary productions do not: someone responsible for the family as a whole, who maintains the variant matrix, runs its checks, and has authority to require changes in one realization for the sake of another. On a conventional set, the continuity supervisor keeps one film consistent with itself across shots. A family needs something like a continuity supervisor for the relations between films.

It also needs performers who understand what they are doing. An actor playing the same blocking twice, once for a thriller and once for a satire, is doing something closer to theater repertory than to ordinary screen acting, and one playing a line so that a single delivery reads two ways under two pictures is doing something that has no established name. The labor of variant production is real and specific, and Chapter 22 argues that it must be credited as such rather than folded into the apparent unity of a single title.

Chapter 16

Editing Under a Common Clock

An editor shapes the rhythm of a film: how long each shot lasts, where the cuts fall, how the pace rises and falls. In a variant family the editor shapes several rhythms at once, each of which must feel right on its own and all of which must stay tied to one clock. This chapter describes that task and the outputs it produces, which go beyond a finished cut to include most of the information the projector and glasses need.

16.1 Temporal debt

Each realization has its own rhythm, and it will not match the spine second by second. Write $\delta_k(t)$ for how far realization k is ahead of or behind the spine at clock time t , measured in story position: how much of the spine's content the realization has reached. A realization that lingers on a reaction falls behind, and δ_k goes negative. One that cuts briskly pulls ahead.

A realization that falls behind has borrowed time, and it must repay it: by cutting something later, compressing a transition, or skipping material the other realizations include. Call $\delta_k(t)$ the realization's *temporal debt*. The envelopes of Chapter 8 require the debt to be cleared at every scene boundary. Stricter synchronization points require it to be cleared within a tolerance ε at particular moments:

$$|\delta_k(t_i)| \leq \varepsilon \quad \text{at each synchronization point } t_i. \quad (16.1)$$

The tolerance depends on what the point synchronizes. A shared musical downbeat or sound effect needs a few frames at most. A structural rendezvous needs only that the realizations have established the same things, and a debt of a second or two does not matter if both have finished the relevant action. Between synchronization points the debt is free, and that freedom is where each realization's rhythm lives.

16.2 The shared track as clock

Under shared or complementary sound, the soundtrack is common to every realization, and it becomes the family's strictest clock. Every audible event in the shared track happens at one moment for everyone. A door slamming, a gunshot, a chord that resolves: each must be compatible with the picture of every realization at that moment.

This is a strong constraint, and a useful one. It means the editor of a shared-sound family works against a fixed timeline of sound events that no realization can move. It also means that the score, in particular, acts as the family's metronome. A composer writing a complementary score, as in Chapter 11, fixes the moments of tension and release for all realizations at once, and the editors of each realization cut to them. The composer and the editors must work together from early on, not in the usual sequence in which music is added after the cut is locked.

16.3 Cuts as hiding places

Several changes in a selective screening are visible if they happen in the middle of a shot and invisible if they happen at a cut. Chapter 3 described changes in slot allocation between streams, which alter each stream's brightness and flash rate. Chapter 2 described adaptive gating, which opens the glasses fully when the streams coincide and closes them to a fraction when they diverge. Both produce steps in brightness. At a cut, the eye expects a change and accepts it. Within a continuous shot, a sudden brightening or dimming looks like a fault.

The editor therefore schedules these changes. Divergence and convergence are placed at cuts where possible, and where a divergence must begin mid-shot, the allocation change is ramped gradually rather than stepped. The same applies to switching. Chapter 19 will discuss whether viewers should be able to switch at any moment. Where switching is allowed, cuts are the natural places for it, because a switch at a cut looks like an edit and a switch mid-shot looks like a jump.

16.4 What the editor delivers

A conventional editor delivers a finished film. The editor of a variant family delivers that for each realization and, in addition, a set of data that the exhibition system requires.

The *display schedule* of Chapter 1: which stream occupies each slot, and how the allocation changes over time.

The *coincidence map*: for each moment, which streams coincide. Following Chapter 2, coincidence is judged on the images as displayed, within a declared perceptual tolerance, not on stored values. It drives adaptive gating and the brightness changes that go with it.

The *switch system* of Chapter 6: its reachable states and admissible transitions over time, computed from the variant matrix and checked against the final cut, since editing can change what is established and when.

The *protection profile*: how strict the leakage requirements are at each moment. Chapter 10 argued that tolerance for leakage should tighten as a family approaches information that one realization must keep from another, and the profile tells the projector when to lengthen blanking intervals or reduce the number of active streams to protect a revelation.

The *composite*, if the family designs one: the unaided view, checked for what it shows at each moment, including what it might reveal.

These deliverables are as much a part of the work as the pictures. A family exhibited without them can be shown, but badly: with wrong brightness, closed glasses during shared passages, unprotected revelations, and switches that land viewers in incoherence.

16.5 Review

Finally, the editor of a family must review it in ways an ordinary editor does not. Each realization must be watched on its own and judged as a complete film. The composite must be watched without glasses. Switches must be tried: at the points the switch system allows, to confirm they work, and at points it forbids, to see what goes wrong and whether the prohibition is worth its cost. And the family must be watched as an audience will watch it, several people in one room on different realizations, listening to the shared track and to each other's reactions. Only the last reveals whether the family is one occasion or several films sharing a wall, which is the question the next part of the book takes up.

Part VI

Spectatorship

Chapter 17

Synchronized Difference

The Introduction stated the book’s central claim: collective viewing does not require that everyone see the same thing; it requires that different experiences remain synchronized closely enough to count as one occasion. Parts I through V supplied the means. This chapter returns to the claim and asks what, precisely, an audience shares when its members do not share a picture, and what “closely enough” has turned out to mean.

17.1 Layers of collectivity

Attending a film with others involves several kinds of sharing that ordinary cinema delivers together and that can be separated. Write the collectivity of a screening as

$$\mathcal{C} = \langle P, T, W, A, D \rangle, \quad (17.1)$$

where P is physical co-presence, being in one room; T is temporal synchronization, experiencing the work on one clock; W is a shared world, the same characters, places, and events; A is common anchoring, points at which every experience corresponds; and D is the permitted divergence, how much the experiences are allowed to differ.

Ordinary cinema sets the first four as high as they go and D to zero. Everyone is in the room, on the clock, in the world, and at every point in the same place, because everyone sees the same thing. Streaming reverses much of this: P and T collapse, since people watch alone and at different times, and D may be large, since recommendation and personalization give different people different works. Private head-mounted displays can keep P and even T while making D unlimited and A absent.

Selective cinema occupies a position none of these do. It keeps P and T as high as ordinary cinema, keeps W and A high by design, and makes D positive but bounded. The audience is neither a mass that sees one thing nor a scatter of individuals seeing unrelated things. It is a *coordinated plurality*: people in one place, on one clock, in one story, seeing it differently.

17.2 What is shared

It is worth listing what the members of a selective audience actually share, because the list is longer than it first appears.

They share the room, its darkness, its seats, its exits. They share the clock: the film begins and ends for everyone at once, and its scenes turn together. They share the world of the story, its people and places and principal events. They share the rendezvous, the moments at which, as Chapter 6 defined, their realizations agree on everything that will matter. Under shared or complementary sound, they share every note of the score and every sound effect, as Chapter 11 described. They share whatever the realizations have in common on screen, which adaptive gating makes literally identical light. They share the composite, the view any of them sees on lifting the glasses.

And they share each other's reactions. The laughter from the comedy's viewers reaches the drama's viewers; the silence of the drama's viewers reaches the comedy's. Each viewer hears, during the film, evidence that others are having a different experience of the same event. This is not a defect to be minimized, though it must be kept from cruelty. It is the audible form of synchronized difference, and it is something no other arrangement produces. Private screens hide the difference. A common screen abolishes it. Selective cinema lets it be heard.

17.3 When is it one occasion?

The book's central claim can now be made more exact. The earlier chapters identified the conditions under which a family holds together, and they can be gathered into a definition.

Definition 17.1 (Occasion-preserving family). A variant family, as exhibited, *preserves the occasion* when three conditions hold:

1. *bounded drift*: every realization's temporal debt is cleared at declared synchronization points, so no viewer's film runs away from the common clock;
2. *recurring agreement*: structural rendezvous recur, so that fact distance is periodically brought back to zero and the realizations repeatedly agree on what has happened; and
3. *a common channel*: at least one element of the experience is received identically by everyone present, such as a shared score, shared footage, or a designed composite.

Each condition answers a specific way of losing the occasion. Without bounded drift, viewers are in the same room at different points in the story, and the room's shared moments no longer line up. Without recurring agreement, the realizations become different stories that happen to share characters. Without a common channel, the viewers are private audiences who share only a wall.

The definition also explains, retrospectively, several of the book's recommendations. Envelopes (Chapter 8) and debt accounting (Chapter 16) secure the first condition. The switch system and rendezvous (Chapter 6) secure the second. Complementary sound (Chapter 11) and adaptive gating secure the third. And the definition makes a prediction: a family exhibited with private sound, long unreconciled divergences, and loose timing will not feel like one screening, however good its optics.

17.4 Against the feed

Synchronized difference has an obvious modern counterpart, and the contrast is instructive. Algorithmic feeds, targeted advertising, personalized search, and recommendation systems

all show different people different things, a condition that has been described as each person living inside a personalized “filter bubble” [11]. They produce divergence without synchronization. Each person’s experience is private, in time and in place, and the divergence is largely invisible: nobody sees the feed of the person next to them, and few know how their own was chosen.

Selective cinema produces divergence under the opposite conditions. It is co-present, synchronized, and, at least in principle, visible. The viewers know that others are seeing something else; they can hear it happening; they can discover afterward what it was. The divergence is declared rather than hidden.

That contrast is the form’s critical potential. It stages, in the oldest social setting cinema has, a condition that contemporary media produce everywhere without showing it: people facing the same surface and receiving different realities. Staged openly, the condition becomes something an audience can notice, discuss, and judge. It also has the opposite potential. The same apparatus could produce divergence that is co-present but hidden, with viewers believing they saw one film. Part VIII is concerned with keeping the first potential and preventing the second.

17.5 Beyond cinema

The structure is not confined to film. A lecture could run several explanatory streams on one screen for students at different levels, all in the room together. A museum installation could show visitors different histories of the same object. A multilingual performance could carry each language on its own stream. A civic presentation could show the same proposal from the perspective of different groups it affects, to an audience that knows each member is seeing one of several. Collaborative work around a shared display could give each participant their own layer.

Cinema remains the clearest case because it has made common exposure central to what attendance means. To go to the cinema is, conventionally, to see what everyone else in the room sees. Selective cinema separates attendance from exposure and shows that the first survives without the second, provided the synchronization is kept. Whether the same holds in lectures, galleries, and assemblies is a question for those settings. The form of the answer, bounded drift, recurring agreement, and a common channel, should carry over.

Chapter 18

After the Screening

A film continues after it ends, in what its audience says about it. In ordinary cinema that conversation starts from a common object. People disagree about what a film meant, what a character intended, whether the ending worked, but they agree about what was shown. In selective cinema that agreement is gone, and the conversation changes character. This chapter examines what happens after a selective screening, and then turns to the protection of what one realization must keep from another's viewers.

18.1 Which one did you see?

After an ordinary film, "What did you think happened?" is a question about interpretation. After a selective one, it cannot be separated from a prior question: "Which version did you see?" Two people may disagree because they read the same evidence differently, or because they were shown different evidence. Until they know which, they cannot tell whether they are disagreeing at all.

The effect depends on the family. After a genre pair, the evidence is nearly the same and the conversation is about tone: one found it frightening, the other funny, and each learns what the other's film made of the same events. After a depth family, the conversation is about understanding, and the viewer of the expert realization may explain what the beginner realization spelled out, or be surprised by how much it did. After a family of endings, the conversation begins with incompatible outcomes, and the audience must reconstruct the family together from its parts. Chapter 10 called this differentiated memory: people who shared a room leave with memories of the same film that cannot both be true of one sequence.

18.2 Three regimes of knowledge

What the audience knows about the variation, before and after, shapes all of this. Three regimes are possible.

In the first, selections are *declared*. Everyone knows the family exists, what its realizations are, and who chose which; the dials are visible, or the choice is announced. The conversation afterward is informed from its first sentence.

In the second, selections are *private*. Everyone knows the family exists and what its realizations are, but each viewer's choice is their own unless they share it. The conversation afterward

includes the choice of whether to say.

In the third, the variation is *undisclosed*. Viewers do not know that others are seeing something different. They discover it, if they do, only in conversation, when a companion describes a scene they do not recognize.

The first two regimes are legitimate and offer different experiences. Declared selections make the family a public game; private selections protect the viewer's choice, which may say something about them, as Chapter 9 noted of expertise. The third regime is different in kind. It turns a work about difference into an instrument of deception: people believe they attended one event and saw one film, and they did not. Some artistic uses might want the shock of discovery, and a work could reasonably withhold which realization a viewer will receive. But the existence of variation should never be concealed. A viewer is entitled to know that the people beside them may be seeing something else. Chapter 24 develops this into a disclosure requirement.

18.3 The conversation as final act

Under the declared and private regimes, the conversation afterward is not an accident of the form. It can be designed for. A family whose realizations each reveal part of a larger picture invites its audience to assemble it. A mystery whose realizations follow different investigators can leave each viewer with a piece of the solution that only the group can complete. A family of endings can be written so that the endings, compared, reveal something none shows alone.

The strongest uses of selective cinema may treat the conversation after the screening as the work's final act: a part performed by the audience, with material the film distributed among them. That is not possible in any form where everyone receives the same thing, and it is only awkwardly possible in forms where people experience a work separately and at different times. It requires exactly what selective cinema provides: people who were together, on one clock, in one story, and were shown different parts of it.

18.4 Protected variation

The conversation depends on something the apparatus cannot fully guarantee: that each viewer actually saw only their own realization. Chapter 2 showed that leakage between streams can be reduced but not eliminated, and that it has two measures, optical and informational. This section turns that observation into a standard.

Not all variation needs protection. A genre pair whose realizations show the same events with different performances loses nothing if a viewer glimpses the other performance for a fraction of a second. Call such variation *casual*. A depth family whose expert realization shows a formula the beginner realization explains later loses a little if the formula leaks early; call it *sensitive*. A family whose realizations withhold a revelation from each other, the identity of a culprit, an ending, a death, loses everything if the revelation leaks; call it *protected*.

The harm of a leak is roughly the product of two factors: how much the leaked information matters, and how recognizable the leak is. A faint luminance ghost carrying no recognizable content is harmless in any realization. A clear glimpse of a face is harmless in a casual family and ruinous if the face is the protected revelation. A standard for leakage should therefore be set by narrative consequence, not by optics alone. Casual variation tolerates the leakage of

ordinary shutter glasses. Sensitive variation needs leakage low enough that no leaked frame is legible. Protected variation needs leakage below legibility, protection of the composite, and particular attention to the moments immediately before and after the revelation.

The protection profile that Chapter 16 made part of the editor's deliverables is the implementation. It tells the exhibition system when to lengthen blanking intervals, reduce the number of active streams, or suppress the composite, so that the strict standard applies only where it is needed and its costs in light are paid only there.

18.5 What protection cannot do

Protection by timing has the limits already identified in Chapter 4. It protects against the eye and not against cameras. A viewer determined to see another realization can adjust their glasses, borrow a companion's, or film the screen. Protected variation in a selective screening is protection against accident, not against intent.

That is enough for its purpose. The point of protecting a revelation is to let viewers who chose not to see it be surprised by it, or never meet it at all. It is not to prevent anyone from ever finding out. The conversation afterward will reveal it in any case, if the viewers want it revealed. Protection keeps the revelation for the people who chose it, at the moment the film intended, and leaves what happens next to the audience. A system that promised more, that a realization could be kept from people who wanted to see it, would be offering concealment, and Chapter 23 examines what that requires and why it changes what the glasses are.

Chapter 19

Choice Before, During, and After

Selective cinema is often described as giving the viewer a choice, and the description hides several different arrangements. A viewer may choose once, before the film, or repeatedly during it. The choice may be the viewer's own, delegated to a companion, left to chance, or made by a system on the viewer's behalf. Each arrangement produces a different work and a different relation between the work and the person watching it. This chapter distinguishes them and argues against the assumption that more choice is always better.

19.1 Preselection

The simplest arrangement is a choice made before the film begins and kept throughout. The viewer sets the dial, perhaps on the advice of a description of each realization, and watches one realization from start to finish.

Preselection asks nothing of the switch system, because nobody switches. It gives each viewer a complete, coherent film. It makes the choice deliberate, a statement of preference made with some knowledge of the options. Its limitation is that the choice is made before the viewer knows what the film is like. A viewer who chose the thriller and finds it too much, or the beginner realization and finds it too slow, is held to a decision made in ignorance.

19.2 Switching during the film

Allowing viewers to switch during the film turns the family into something that can be explored. A viewer can sample realizations, move toward more explanation during a hard stretch, or follow a character who has become more interesting. The family stops being a set of films and becomes a space the viewer moves through.

Switching is also where the switch system of Chapter 6 does its work. A viewer who switches carries a history, and not every switch keeps that history coherent. The exhibition has three broad options. It can *lock* the dial except for admissible switches, so that the viewer can only make moves the family supports. It can *warn*, allowing any switch but signaling when a move will leave the viewer without something they need or contradict something they were shown. Or it can restrict switching to *rendezvous*, where, for viewers who have not yet switched, every move is admissible and nothing needs to be tracked.

Each option expresses a different view of the viewer. Locking treats the family's coherence as the artist's responsibility and protects the viewer from incoherence they did not intend. Warning treats the viewer as entitled to make incoherent moves knowingly, perhaps out of curiosity about what the other realization is doing. Rendezvous-only switching is the simplest, needs no state in the glasses, and gives the film a rhythm of choice points like chapter breaks.

19.3 The artist's right to constrain

It is tempting to regard any restriction on switching as a limitation to be minimized, and to treat the ideal family as one in which every viewer can move anywhere at any time. The book does not take that view.

A novel does not let its reader substitute paragraphs. A symphony does not let its listeners change the order of movements. These are not defects. They are part of what makes the work a work: a designed sequence, whose effects depend on its order. Music offers a closer precedent than either. Eco described "open works" whose performers or readers complete them within limits the author sets, such as Stockhausen's *Klavierstück XI*, whose pianist chooses the order of its fragments during the performance [4]. The freedom in such works is real and it is composed. A variant family is a designed space of the same kind rather than a designed sequence, but it is still designed, and its effects can depend on where and when a viewer is allowed to move. A mystery whose realizations follow different investigators may depend on each viewer staying with one investigator until the solution. A family of endings may depend on the choice of ending being made at the branch and not before.

An artist who restricts switching is therefore not failing to offer choice. They are deciding what kind of work to make. The switch system makes such restrictions principled rather than arbitrary: they can be tied to coherence, allowing moves that keep the viewer's history intact and forbidding those that would break it, rather than simply closing the dial. But even a restriction that goes beyond coherence, keeping a viewer on one realization for reasons of form, is a legitimate artistic choice, provided it is declared. A viewer should know, before choosing, whether they will be able to change their mind.

19.4 Delegation

A viewer may let someone else choose. A parent may choose for a child, a friend may choose for a friend, a teacher for a class. Delegation is common and often reasonable. A young child may not be able to read a description of the realizations, and a parent's choice is the ordinary way such decisions are made.

It is worth noticing how much the complementary design of Chapter 8 eases delegation. When the realizations are the thriller and the children's satire of the *Protocols* pair, a parent choosing the satire for a child is choosing an address, not guarding against harm, and the stakes of a wrong choice are low. When the realizations include one that a child should not see, delegation carries a burden the glasses cannot share, because, as Chapter 4 showed, the child can remove the glasses and see the composite. Delegation in a restrictive family is a matter of trust and supervision, not of technology.

19.5 Random assignment

A screening can assign realizations at random: each ticket, each pair of glasses, each seat, set to a realization nobody chose. This makes variation part of the event. The audience knows the family exists and does not know which realization each of them will receive, and the conversation afterward begins from genuine discovery.

Random assignment is a legitimate artistic device, with one condition carried over from Chapter 18: the existence of variation must be disclosed. Viewers can agree to be surprised by which realization they receive. They cannot meaningfully agree to a variation they do not know is happening.

19.6 Adaptive selection

The last arrangement is the one this book most resists. A system could choose the realization for the viewer, and change it during the film, on the basis of what it observes: where the viewer looks, how they react, how their heart rate changes, what they have watched before, what is known about them. Such a system would promise a film continuously tailored to its viewer.

Adaptive selection requires the glasses, or something in the room, to observe the viewer. It would extend into the cinema the practice, well documented in digital services, of collecting behavioral data in order to predict and shape what people do [16]. It abandons the architectural guarantee of Chapter 2, that the simplest glasses cannot record or measure anything, and replaces it with a sensor pointed at a person. It moves the choice from the viewer to a system whose reasoning the viewer cannot see. And it makes the question Chapter 9 raised about expertise, who decides what a viewer can follow, into a question answered continuously by inference, without the viewer's participation.

None of this is inherent in selective cinema. The glasses of Part I are timing devices. They can be set by hand, and they need to know nothing about the person wearing them. Adaptive selection is an addition, and it should be treated as one: never the default, always declared, and always refusable in favor of a realization the viewer chooses. Chapter 24 returns to the specific danger of a system that observes viewers and then chooses what they see.

19.7 Choosing to choose

The arrangements in this chapter are not exclusive, and a single screening can combine them: a preselected realization with switching allowed at rendezvous, or random assignment with a warning dial for the curious. What matters is that the arrangement is part of the work and is known to the audience. A viewer entering a selective screening should know three things before it begins: what the realizations are, how their own will be chosen, and whether they will be able to change it. With those three things known, choice before, during, and after the film becomes one more element an artist can compose, and one the audience can understand.

Part VII

Access and Adaptation

Chapter 20

Accessibility Without Peripheralization

Accessibility in cinema has usually been delivered at the edges: captions at the bottom of the frame, a sign-language interpreter in a corner or a separate screening, audio description through headphones, a device in the cupholder. The accommodation is real and the placement says something. The film is for the typical viewer, and access is attached to it. Selective cinema offers a different arrangement, in which an accessible realization occupies the whole screen as a member of the family. This chapter considers what that makes possible and what it obliges.

20.1 Access streams

Several kinds of accommodation can be carried as streams.

A *sign-language realization* can place the signing performer within the composition rather than beside it: framed as part of the scene, or as a character's inner voice, with the film's images composed around the signing rather than shrunk to make room for it.

A *captioned realization* can carry text designed as part of the image, placed where it reads best in each shot, rather than confined to a fixed band.

A *low-intensity realization* can reduce flashing, rapid cutting, sudden loudness in the picture, and visual clutter for viewers with photosensitivity, vestibular sensitivity, or sensory overload, as an intensity transformation in the sense of Chapter 7.

A *focal realization* can enlarge the action that matters, simplify the composition, and hold shots longer for viewers with low vision or slower visual processing.

What these share is that they are visual. That matters, because Chapter 11 showed that visual variation fits naturally with shared sound. A signing realization, a captioned realization, and a focal realization can all run under the same complementary score, and their viewers remain part of the room's shared audio. Accommodations that are carried by sound, such as audio description, need individual sound and carry its costs, but even these can be designed with the family rather than added afterward.

20.2 Parity

Carrying access on the main screen does not by itself make it equal. A system that can run several streams can still give the accessible one the least: the weakest writing, the least rehearsed

performance, the lowest share of light, the fewest distinct frames. The danger is peripheralization moved from the edge of the frame into the schedule.

The obligation this chapter proposes is *parity*. An accessible realization should be judged as a complete film, by the same standards as the others, not by whether it transmits the minimum information a viewer needs to follow the plot. A signing realization should be composed by people who understand sign language as a visual art, not produced by pasting an interpreter into existing footage. A focal realization should be edited for its viewers, not cropped from someone else's cut.

Parity has a concrete meaning in the terms of Part I. Every stream in a family receives some share of light and flash rate. Parity requires that access streams receive the same share as the others under the same conditions, and that the family's protection profile and switching apply to them equally. An access stream that is dimmer, lower in frame rate, or excluded from switching is peripheral however central its position on the screen.

20.3 Competition for capacity

Parity creates a hard choice that should be faced openly. Chapter 3 showed that a present-day projector can carry two or three streams. If one of them is an access stream, it is not available for a second genre, a second depth, or a second ending. Access competes with artistic variation for the same small budget.

There are three responses. The first is to use the budget efficiently. Access realizations often share most of their frames with another realization: a captioned realization differs from its uncaptioned counterpart only where text appears, and adaptive gating keeps the glasses open everywhere else. Such access streams cost little light. The second is to design access into the family rather than beside it. A family whose realizations are the complementary genre pair of Chapter 8 could caption both, or make one of them the focal realization, so that access is a property of existing realizations rather than a stream of its own. The third is to decide, and to say, that access comes first when a choice must be made. The book takes that position, on the ground that an artistic dimension that excludes some viewers is worth less than one that includes them.

20.4 The glasses themselves

Selective cinema has an access problem of its own that should not be hidden. The glasses must be worn. Some viewers cannot wear them comfortably, some wear prescription lenses that make fitting harder, and the flicker of the shutters, though well above the rates most associated with photosensitive seizures, is not tolerated by everyone. Adaptive gating adds a further consideration: its brightness steps, if they recurred rhythmically at low frequencies, could produce exactly the kind of flashing that photosensitive viewers must avoid. Editors should never schedule gating changes as a regular pulse, and the protection profile should include limits on how often brightness may change.

For viewers who cannot use the glasses at all, the composite of Chapter 4 is what they see. In coincident passages it is the film, correct and at full brightness. In divergent passages it is whatever the family made of it. A family that designs its composite gives these viewers a real realization; one that ignores it gives them a double exposure. Designing the composite

is therefore partly an access decision, and a family that cares about access should treat the unaided view as a realization owed to the people who have no other.

20.5 Access as a design dimension

The chapter's argument reverses the usual order. Accessibility is usually something done to a finished film. In a variant family it can be one of the dimensions the family is built on from the start, with its own column in the variant matrix, its own checks, and its own claim on the budget. That is more demanding than adding captions afterward, and it is the only way to give access the central position selective cinema makes available rather than rebuilding the old periphery inside the new schedule.

Chapter 21

Instruction and Translation

Two institutional settings make especially strong use of selective presentation: teaching and translation. In both, a room of people with different needs faces one presentation, and in both, the usual remedies divide the room or slow it to a single pace. This chapter considers what selective streams could do in classrooms and multilingual settings, and where they could do harm.

21.1 One demonstration, several explanations

A shared demonstration, an experiment, a procedure, a worked problem, can be shown to a whole class while each student receives the explanation they need. The depth family of Chapter 9 is the natural structure: introductory, intermediate, and advanced realizations of one demonstration, synchronized, each complete, with the one-way window allowing any student to move toward more explanation whenever they need it.

The educational value is plain. A single presentation normally forces one pace on everyone, too fast for some and too slow for others. A depth family lets each student follow at their level while remaining in the same room, watching the same demonstration, discussing it afterward with people who saw the same events explained differently. The classroom remains a group.

21.2 Temporary scaffolding

The most useful feature for teaching may be the least conspicuous. A student need not choose a level once. They can remain on the common or advanced realization and step into a more explanatory one for a single difficult interval, returning when they have what they need. The one-way window guarantees that the step toward more explanation is always admissible, and the return becomes admissible once the explanation is complete.

This is scaffolding in the ordinary teaching sense: support taken when it is needed and set aside when it is not, without the student leaving the room, asking in front of others, or being moved to a different group. It is available only because the explanation happens in time the common screen would otherwise have to share, and only because the switch system keeps the student's history coherent when they move.

21.3 Selection and tracking

The same system can be used very differently. An institution could assign students to realizations: this group to the introductory stream, that one to the advanced, on the basis of test scores, prior grades, or a teacher's judgment. That is tracking, whose effects on students assigned to lower tracks have been extensively documented [10], and selective presentation would make it invisible: students in one room, apparently receiving one lesson, actually sorted into levels they did not choose and may not know about.

Chapter 9 argued that depth should be a property of the presentation a viewer selects, not a rank assigned to the viewer. In a classroom the principle has specific consequences. Students should choose their own realization, with advice if they want it. Their choice should be private: a student's dial position is not the teacher's business unless the student makes it so. They should be able to change it at any time the switch system allows. And nothing that follows should depend on which realization they watched. An assessment that rewards the advanced realization's content penalizes students for choosing help, and turns a tool for learning into a sorting device.

These principles will not always be kept, and the risk should be stated clearly. Selective presentation in schools could make tracking more efficient and less visible than it has ever been. The same apparatus that lets a student take help privately lets an institution withhold material privately. Which it does depends on who controls the dial.

21.4 Translation in the room

Chapter 13 discussed language as a dimension of variation between theaters. In a single multilingual room, the same principles apply at smaller scale. Subtitle streams in several languages cost little light, because they differ only in their text. Explanations can be localized as well as translated, using examples that make sense to each group. Under individual sound, spoken language can vary; under shared sound, the spoken language is common and the others are carried in the picture.

A multilingual classroom or public meeting could thus share one presentation, one room, and one clock while each participant receives it in their own language. The shared score, the shared images, and the rendezvous hold the group together; the language streams let each person take part.

21.5 Plural authorship

Translation at this depth raises a question Chapter 13 began. When a realization is not only translated but localized, with its examples, jokes, and references rewritten, the translator is an author of that realization. In an ordinary film, translation is credited as a service to the original. In a variant family designed with its language realizations from the start, there may be no single original, and each realization's writers are its authors. Credit and provenance should say so: which realization was written by whom, from what source, with what changes. Chapter 22 develops this into a general requirement.

Part VIII

Control of the Variant Set

Chapter 22

Who Defines the Versions

Once a work exists as a family of realizations, control over the family becomes a form of power. Someone decides which realizations exist, which are shown in a given place, who may see which, and at what price. In ordinary cinema these decisions are dispersed and mostly visible: a studio releases a cut, a censor classifies it, a theater programs it. In selective cinema they can be made inside a single screening, and they can be made invisibly. This chapter asks who should make them.

22.1 Authorial and distributive variation

It helps to distinguish two sources of variation. *Authorial* variation is created by the makers of the work as part of its design: a director who makes comic and dramatic realizations, writers who compose a family of endings. *Distributive* variation is introduced after the work is made, by those who handle it: a distributor who replaces a sequence for a particular market, a theater that disables a realization, a sponsor who pays for a stream with altered product placement, a regulator who requires a cut.

Authorial variation is part of the work. Distributive variation changes it, and it may do so in ways the makers never intended and the audience never learns about. The apparatus cannot tell the two apart; a stream is a stream. The difference lies in who introduced the variation and whether its introduction is declared.

A selective work should therefore carry its provenance. Which realizations were made by its authors, which were added or altered afterward, by whom, and why. A realization the authors did not make is not thereby illegitimate, since local realization in Chapter 13 is often exactly this, but it should not be presented as the authors' work.

22.2 Assignment by category

The earliest illustration of selective cinema was an adult and a children's version on one screen, and it suggests assigning realizations by category: by age, and by extension by language, ticket class, region, disability, subscription tier, or inferred sensitivity. Some of these are reasonable. A viewer who needs a language will want the realization in it. Others amount to sorting people without telling them.

The book's position, developed across Chapters 9, 19 and 21, is that silent assignment should not be the default. Viewers should be able to see what dimensions a family varies along, choose among them, and know when a choice has been made for them and on what basis. "The version suitable for you" is not an adequate description when the grounds of suitability are withheld. A viewer assigned a realization because of a category they belong to, or one a system inferred, is owed the knowledge that this happened.

22.3 Price

A family's realizations could be priced differently. The expert realization might cost more, a preferred ending might be sold as an upgrade, a local performance offered at a premium. Some of this may reflect real costs. Much of it would be artificial scarcity. Once several realizations are already being projected in a screening, every viewer's glasses could admit any of them; disabling one for a cheaper ticket is a decision about access, not a technical necessity.

Tiered pricing inside a single room also produces something ordinary cinema rarely does: visible class difference within one audience's experience of one work. Two people in adjacent seats, one of whom paid to see the ending and one of whom did not, are not sharing an occasion in the sense of Chapter 17. They are receiving a stratified product. A form whose promise is shared presence with different experiences should be wary of turning the difference into a price list.

22.4 Honest description

Chapter 15 observed that a family's count of realizations is a poor measure of what it offers. Three dimensions of two options each make eight nominal realizations, whether they differ greatly or hardly at all. Advertising "eight films in one" when four of the eight differ only in subtitles misleads, and the temptation to inflate counts will be strong once the format is sold on them.

Honest description names the dimensions and says how far apart the realizations are along each. A family might be described as two genres of one story, sharing all events and differing in performance and score, with two endings that diverge in the final ten minutes, and captioned throughout. That is longer than a number and much more useful. It tells a viewer what they are choosing among.

22.5 Labor and consent

A variant family is made by more people doing more kinds of work than its single title suggests: writers of each column and of the spine, performers who play the same blocking in several registers, composers of a complementary score, editors balancing temporal debt, the continuity supervisor for the family proposed in Chapter 15, translators and local performers, the makers of access realizations. Credit should record who made each realization and who designed the architecture that holds them together. Otherwise the apparent unity of the work erases the plurality of its makers.

Performers raise a specific issue. An actor who agrees to play a scene in a thriller has not thereby agreed to have the performance appear in a satire, recut into a different context, or

re-rendered to speak lines in another language. The flexibility that makes variant families possible, modular footage, re-editing, synthetic re-rendering, makes it easy to use a performance in realizations the performer never approved. Consent should be given per realization, and a performance should not appear in a realization, or a composite, that its performer has not agreed to.

22.6 Control of the dial

All of these questions reduce to one: who controls the dial? In the design this book has argued for, the viewer does. The glasses are timing devices set by hand; the realizations are declared; the variation is authorial or its provenance is stated; assignment, when it happens, is disclosed. Every departure from that design moves control from the viewer to someone else. Some departures are justified. None should be made silently. The next two chapters examine the two most consequential departures: glasses that only licensed viewers can use, and variation used to tell different audiences different things.

Chapter 23

Perceptual Rights Management

Among the early notes behind this book is a proposal that sits uneasily with the rest of it: blink-rate glasses with patented algorithmic sequences, so that certain works could be seen only by glasses set to the correct sequence, and “blink-rate protectors” that let only authorized frames be seen. The same conversation discussed Cory Doctorow, one of the most persistent critics of digital rights management. This chapter takes the proposal seriously, as a design and as a warning, because it marks the point at which a device for selecting what one sees becomes a device for controlling it.

23.1 From phase to key

Chapter 2 ended with a distinction. Periodic gating, in which each stream occupies a fixed phase of a fixed period, makes a film selectable: anyone with adjustable glasses can see any stream. Concealment requires something else. The natural candidate is a schedule that is not periodic, in which slots are assigned to streams by a pseudorandom sequence generated from a secret key. Glasses holding the key open at the right moments and see the stream. Glasses without it open at the wrong moments and see a scramble of fragments from every stream, or nothing coherent at all.

This is technically plausible. It is also a change in what the glasses are. Periodic glasses need only a timing reference and a dial. Keyed glasses need a secret, a way to receive it, a way to keep it from their owner, and usually a way for the exhibitor to confirm that the glasses are approved hardware that will not leak the key or the stream. They become, in the vocabulary of rights management, a trusted device: trusted, that is, by the licensor, and trusted precisely because they do not do what their wearer might want.

23.2 Rights management for perception

Digital rights management restricts what people can do with copies of works: whether they can copy, lend, move them between devices, or keep them after a subscription ends. Doctorow’s criticism, sustained over many years [3], is that such systems do little to stop determined copying while imposing real costs on ordinary users, locking them to vendors, making them unable to repair or inspect their own devices, and, where the law forbids circumven-

tion, as section 1201 of the United States Digital Millennium Copyright Act does, making it an offense to find out what their own equipment is doing.

Keyed gating would move rights management from the copy to the act of seeing. What is restricted is no longer a file but a percept: whether a person in the room, looking at the screen, may see what is on it. The restriction would be enforced by a device on the viewer's face that they could not inspect or modify without breaking it, and perhaps without breaking the law.

The consequences follow from the device's new requirements. A trusted device must be authenticated, and authentication tends to require identity: which glasses, licensed to whom, for which work. The non-recording guarantee of Chapter 2, that the simplest glasses can be shown by inspection to measure nothing, is lost the moment the glasses must hold a secret and prove their legitimacy. The glasses must now communicate with an authority, and a channel that carries a key to the glasses can carry information from them.

23.3 What keys cannot do

Keyed gating also fails at its own purpose, for a reason Chapter 4 identified. The key controls when the glasses open. It does not control the screen. Every slot of every stream is shown in the room, to anyone who looks, and a camera with a short exposure, triggered at the right moments or simply run at a high frame rate, records the slots directly. Separating the streams afterward is a matter of finding the pattern, which a keyed schedule makes harder but does not make impossible, and which becomes trivial if one pair of licensed glasses is used to reveal it. This is a version of what rights-management critics call the analog hole: whatever is presented to a human sense can be captured by an instrument pointed at the same place.

So keyed gating protects against the unaided eye and against casual glasses, and fails against anyone who brings a camera. It costs viewers the inspectability of their own equipment and protects works only against those least likely to copy them. That is the characteristic trade of rights management generally, now moved onto the face.

23.4 The other proposal

The same notes contained a different proposal, and it points somewhere better. A visitor is scanned, as at an airport, to show they carry no cameras or recorders, and then enters a room alone, where they are shown an original work made for them. On leaving, they receive a spoken passphrase that will let them see the work again later.

This arrangement protects the work without putting anything on the viewer. The protection is the room: the absence of recording instruments, secured by a check at the door. The access afterward is a phrase held in the viewer's memory rather than a key held in a device. Nothing the viewer wears or carries is made untrustworthy to them. The work is protected by presence and by agreement, the conditions under which it is shown, rather than by an instrument that restricts what the viewer's own eyes can see.

The contrast is instructive. Keyed glasses try to protect a work while it is shown to a room that may contain cameras, and they fail against the cameras while costing the viewers. The private room removes the cameras, and then needs no keys. It is also honest about what it is: a restricted viewing, entered knowingly, under stated conditions. It does not pretend to be a public screening.

23.5 A position

The chapter's position has three parts.

First, selective cinema should be built on periodic gating. Its glasses should select and not restrict, carry no secrets, and remain inspectable, and the phases of a family's streams should follow an open standard so that any compliant glasses can use them.

Second, where a work genuinely requires restricted viewing, the restriction should be made by the conditions of viewing, such as a private room or a closed screening, rather than by locking the viewer's equipment. Keyed gating, if used at all, should be confined to works declared as restricted, shown to audiences who know they are entering a restricted viewing, with glasses whose operation is documented.

Third, the distinction between selection and restriction should never be blurred in how the technology is described or sold. A system that lets viewers choose among realizations and a system that prevents viewers from seeing realizations are different systems with different obligations. Calling the second by the name of the first would carry the trust earned by an honest filter into a device built to be trusted by someone else.

The early notes proposed both patented sequences and the private room. The first is the more obvious extension of the glasses. The second is the better idea.

Chapter 24

Differential Persuasion and Disclosure

The gravest misuse of selective cinema does not require restricting anyone's glasses. It requires only that different people be shown different things while believing they saw the same film. This chapter examines that possibility and proposes the disclosure that would prevent it.

24.1 The hazard

A multiplexed film could show different products to different viewers, different political messages, different characterizations of the same public figure, or different moral conclusions to the same story. Each viewer would see a coherent film. The viewers might discuss it afterward, each assuming their film was the film. Critics would review one realization and believe they had reviewed the work. Someone who objected to a message would find that others in the same screening had not seen it.

Differential messaging is already common in advertising and political communication, which target different groups with different content. What selective cinema would add is its appearance of commonality. A film screened in a public room has always functioned as evidence that the audience received one text. That is the assumption that makes public criticism possible: there is an object, and anyone can go and see it. A selective film with undisclosed variation would keep the appearance of a common object while removing the object. Criticism would lose its target, because no accessible version contains all the variations and nobody knows which they saw.

The danger is sharpest when variation is combined with the adaptive selection of Chapter 19: a system that observes each viewer and then chooses what they see. That combination is targeted persuasion presented as a public screening, and it is the use this book most wants to see excluded.

24.2 Disclosure of existence

The first requirement, already stated in Chapter 18, is that the existence of variation be disclosed. A viewer entering a selective screening must be told that the audience may be seeing different realizations. This single requirement removes the worst of the hazard, because it removes the false assumption of commonality. A viewer who knows that others may be seeing something else will not take their own film for the film.

24.3 A variant registry

Disclosure of existence is necessary and not sufficient. A viewer who knows variation exists still needs to know what it consists of. The second requirement is a *variant registry*: a public record, for each selective work, of what realizations exist, what dimensions they vary along, what each contains, and who made each. The registry is the public version of the variant matrix of Chapter 14, or of the parts of it that concern what viewers see.

The registry makes criticism possible again. A critic can review the family, not a member of it. A viewer who objects to something in their realization can find out whether others saw it. A researcher can compare what was shown to whom.

24.4 Manifests and verification

A registry says what a work contains. It does not prove that what was screened matches it. A distributor could register one set of realizations and project another.

The third requirement is a *manifest*: a record of what each stream contained as exhibited, with a cryptographic fingerprint of each stream's content, published with the registry. Anyone with access to the exhibited material, a regulator, a critic, an archivist, can check that the fingerprints match. A registered realization that differs from what was shown will not match its fingerprint.

The manifest also answers a question each viewer may reasonably ask: which realization did I see? In the design of Chapter 2, the glasses know only their own dial position and the timing signal. A viewer can read their own phase from the glasses and look it up in the public manifest for that screening. Nothing about the viewer is recorded or transmitted, and no one else learns which realization they chose. The viewer learns what they saw without the theater learning who they are.

24.5 Reference screenings and audit

The fourth requirement is that every realization be available for inspection. A selective work should be shown, at least in reference screenings, in a mode in which any realization can be watched in full, including the composite. Archives should hold every realization, as Chapter 25 argues. A work whose realizations cannot all be seen by someone accountable is a work whose content is, in part, secret, and a secret shown in public to different people is the hazard itself.

24.6 Sponsors

Commercial pressure deserves a specific word. A sponsor could pay to have its product appear in one realization, or to have a realization made for an audience it wants to reach. There is nothing inherently wrong with product placement, but its legitimacy in ordinary film rests on its visibility: everyone sees the same placements, and anyone can point them out. Placements that vary by realization, invisibly, are a different practice. They should be listed in the registry like any other variation, with the sponsor named, and no realization should exist primarily to carry a message that a sponsor paid for and the audience was not told about.

24.7 What disclosure preserves

The requirements of this chapter are not burdens on the form. They are what allows it to keep the promise of Chapter 17. Synchronized difference is valuable because it makes divergence visible, co-present, and discussable, the opposite of the hidden divergence of personalized feeds. Undisclosed variation turns the form into a more concentrated version of the thing it could have exposed. Disclosure keeps it on the right side of that line. With it, a selective screening is a public event in which everyone knows that everyone is seeing the story differently. Without it, it is a private message delivered in a public room.

Chapter 25

Which Film Is the Film

The hero of the thriller at the origin of this book is named Theseus. The name recalls an old puzzle. The ship in which Theseus sailed was preserved by the Athenians, who replaced its planks one by one as they rotted, until none of the original timber remained, and philosophers asked whether it was still the same ship [12]. Selective cinema poses a stranger version of the question. It is not that one ship's planks are replaced over time. It is that several ships sail at once, sharing some planks and not others, and all of them carry the same name. Which is the ship?

25.1 The question

The question is not idle. Critics must know what they are reviewing. Archives must know what to preserve. Courts, in disputes about rights, defamation, or contracts, must know what a work contains. Audiences who argue about a film must know whether they are arguing about the same thing. In ordinary cinema the answer is conventional: the work is the released cut, and variants are versions of it. In selective cinema, where several realizations are released together, shown together, and none is necessarily primary, the convention does not apply.

25.2 Four answers

There are four candidate answers.

The first privileges one realization as the *master* and treats the others as variants of it. This is the answer of the primary family in Chapter 5, and for some families it is correct: a work made as a feature with an explanatory version for students has a clear original. But it fails for equal families, where the makers declared no primary realization, and it imposes on those works a hierarchy they were designed to avoid.

The second takes the *family* as the work: the set of all its realizations, each equally part of it. This respects equal families, but a set is not quite enough. Two unrelated films of equal length form a set. What makes a family a work is how its members relate.

The third takes the *generative specification* as the work: the scene model, the rules for realizing it, and the transformations allowed. This fits works made by the virtual production of Chapter 15, where realizations are rendered from a model. It fits less well works shot on film, where the realizations are primary and no model exists apart from them.

The fourth takes the *record of exhibition* as the work: every realization as actually shown, in every venue, at every screening. This is the most complete answer and the least usable. It makes the work unbounded and never finished, since each screening adds to it.

25.3 Identity in constraints

The book's answer combines the second and third. A selective work is identified by its *declared constraints*: the shared world, the realizations authorized, the synchronization that ties them together, the transformations that relate them, and the rules governing how viewers may move among them.

The earlier chapters have already named these constraints. They are the spine and columns of the variant matrix in Chapter 14. They are the envelopes and rendezvous that keep the family synchronized. They are the switch system of Chapter 6, which states which histories can be continued into which realizations. They are the provenance of Chapter 22 and the registry of Chapter 24. Gathered together, they can be written as

$$\mathfrak{F} = \langle \mathcal{M}, \mathcal{V}, \mathcal{A}, \mathcal{S}, \mathcal{L}, \mathcal{E} \rangle, \quad (25.1)$$

where $\mathcal{M}$ is the shared world model, $\mathcal{V}$ the authorized realizations, $\mathcal{A}$ the synchronization anchors, $\mathcal{S}$ the selection and switching rules, $\mathcal{L}$ the transformations linking realizations, and $\mathcal{E}$ the exhibition conditions under which the work is meant to be shown.

A screening is a screening of the work when what it shows is generated within these constraints: realizations from the authorized set, synchronized at the declared anchors, switchable under the declared rules, exhibited under the declared conditions. A realization that violates them, an undeclared distributive alteration, an unauthorized stream, a sponsor's insertion not in the registry, is not the work, however it is titled.

25.4 A precedent

The idea that a work can be identified by its constraints rather than by a single instance has precedents. A chess problem is identified by its position and its stipulation, not by any particular game played from it; many solutions may be attempted, and the problem is the same problem. A piece of music with improvised passages is identified by its structure and its fixed parts, not by any single performance. A play is identified by its text, and every production realizes it differently.

Goodman gave the distinction a name. In his terms, an *allographic* art is one whose works are identified by compliance with a notation, as a symphony is by its score, rather than by the history of a particular object, as a painting is [5]. The variant matrix and switch system are not a notation in Goodman's strict technical sense, but they play a comparable role: they state what any authorized realization of the work must satisfy. Selective cinema moves film toward the allographic side of that line.

Cinema has been unusual in fixing its works to a single sequence of frames. Selective cinema returns it toward the condition of the other arts, in which a work is a structure that admits many realizations and remains itself across them. The Ship of Theseus puzzle arises because a ship seems to be its planks. A selective film is not its frames. It is the design that determines which frames may be shown together, to whom, and in what relation.

25.5 What archives must keep

The answer has practical consequences for preservation. An archive that keeps only one realization has kept a member of the work, not the work. An archive that keeps every realization but not their relations has kept a pile of films. What must be kept is the declared constraints and the realizations together: the variant matrix, the switch system, the registry and manifests, every authorized realization, the designed composite if there is one, the complementary score, and, for works produced from models, the models and the rules for rendering them, with the software needed to run them.

To these should be added representative records of exhibition: what was actually shown at some screenings, with their manifests. These are not the work, but they are evidence of how the work was realized, as recordings of performances are evidence of a piece of music.

25.6 Authorship relocated

The answer also says something about authorship. In ordinary cinema, the authors choose every frame. In selective cinema, they choose a space of frames and the rules for moving through it. Some of the authorship moves from the sequence to the design: from deciding what is seen to deciding what may be seen, in what combinations, by whom.

This does not diminish authorship. The designers of a variant family decide more, not less, than the makers of a single film: each realization, and every relation among them. What it changes is where authorship is visible. A single film displays its choices. A variant family displays some of them to each viewer and keeps the rest in its structure, which only the whole family, and the conversation of an audience that saw different parts of it, can reveal.

Part IX

Evaluation

Chapter 26

Evaluating a Family

A variant family can fail in more ways than a single film. Its optics can leak, its realizations can be incoherent alone or together, its switching can strand viewers, its audience can feel that it attended several events rather than one, and its equipment can do things its wearers never agreed to. This chapter sets out how a family and its exhibition should be evaluated, in four groups of tests: technical, narrative, social, and ethical. Appendix D collects them as a protocol.

26.1 Technical evaluation

The optical system is measured first, because everything else depends on it.

Timing. The display's refresh must be stable, the glasses must hold phase against the synchronization signal, and drift must be measured over a full screening, not a few minutes. Phase error that is harmless at the start of a film can accumulate.

Leakage. Crosstalk is measured in both of the senses Chapter 2 distinguished. The optical measure is the fraction of light from other streams reaching the eye. The informational measure is whether leaked content is legible: whether test viewers can identify faces, words, or objects flashed in other streams at the family's actual leakage level. The informational test is the one that matters for protected variation, and it must be run at the moments the protection profile marks as sensitive, with the blanking and allocation settings used there.

Luminance and flicker. Each stream's luminance is measured through the glasses, in coincident and divergent passages, and flicker is assessed centrally and peripherally at those luminances. Brightness changes from adaptive gating and allocation shifts are logged, and their timing is checked to confirm they fall at cuts or ramp gradually, and never recur as a regular pulse.

Seating. All of these are measured across the auditorium, not only at its center: viewing angle affects some shutter technologies, and distance and screen size affect flicker and legibility.

Failure. Every exhibition must declare what the glasses do when they lose the timing signal or their power. There are three choices. Glasses that go transparent show the composite; if the composite is designed, this is a safe failure, and if the family contains protected variation, it may expose it. Glasses that go opaque interrupt the film but reveal nothing. Glasses that fall back to a declared common stream keep the viewer in the film but may place them in a

realization they did not choose. The right choice depends on the family's protection profile and may change during the film. What is not acceptable is an undeclared failure mode, discovered by an audience during a screening.

26.2 Narrative evaluation

Each realization is watched alone and judged as a complete film. The variant matrix checks of Chapter 14 establish coherence; this evaluation establishes quality, which no check can.

The family is then evaluated as a family. Reviewers confirm that its realizations stand in the intended relations: that the genre pair tells one story, that the depth family's explanations serve their viewers, that the endings are each earned. They look for unintended asymmetry, such as one realization given the better writing or the stronger performances, which is especially important when one realization is an access stream.

Switching is tested directly. Reviewers switch at the points the switch system allows, to confirm that the moves work as intended, and at points it forbids, to see what goes wrong and whether the prohibition is worth what it costs the viewer. They try paths with several switches, since Chapter 6 showed these can fail when single switches do not.

Finally, reviewers watch different realizations in one room and compare their memories afterward. The comparison shows what each realization actually established for its viewers, which may differ from what the matrix says it establishes, for the reason Chapter 9 gave: presentation is not uptake.

26.3 Social evaluation

A family can be technically clean and narratively strong and still fail at what the book has argued is its purpose: to be one occasion experienced differently. That failure can only be found by watching audiences.

The definition of an occasion-preserving family in Chapter 17 gives measurable proxies. Bounded drift can be measured from the editor's data. The frequency of structural rendezvous can be read from the switch system. The presence of a common channel is a fact about the exhibition. But the definition's success is a claim about experience, and it must be tested by asking audiences.

Relevant questions include whether viewers felt they had attended the same event as the others in the room; whether hearing others' reactions to a different realization felt like connection, confusion, or intrusion; whether the conversation afterward was enriched or frustrated by the difference; whether viewers felt ranked or sorted by their choice; whether companions coordinated their choices, and why; and, for families with a designed composite, whether anyone who saw it understood it as part of the work. Preference ratings alone will not answer these questions. A family can be enjoyed by every viewer and still have failed to be one occasion, if each experienced it as a private film projected on a public wall.

26.4 Ethical audit

The last group of tests concerns the commitments of Part VIII, and it is the easiest to omit, because nothing in the film breaks if it is skipped.

The glasses are inspected. The claim of Chapter 2, that the simplest glasses can be shown to record and measure nothing, is only as good as the inspection that confirms it for the glasses actually handed out: no cameras, no eye trackers, no storage beyond the dial state, no transmitter beyond what timing requires.

The registry and manifest of Chapter 24 are checked against what was exhibited, stream by stream.

The provenance of every realization is confirmed: which were made by the work's authors, which were added or altered afterward, and whether the alterations are declared.

The disclosure given to the audience is checked for what it actually says: that variation exists, what the realizations are, how each viewer's realization is chosen, and whether it can be changed.

A family that passes the first three groups of tests and fails this one should not be exhibited. It may be excellent cinema. It is also, in the terms of the previous chapters, an instrument for telling different people different things without their knowledge, and the quality of the cinema makes that worse, not better.

Conclusion: Sharing Time Without Sharing Every Frame

Return to the two people in adjacent seats. They came in together, sat in the dark, and looked at the same screen for an hour and a half. The film began for both at once and ended for both at once. They heard the same music, the same doors slamming, the same laughter from the far side of the room. They followed the same characters through the same events, and at several moments in the film, though neither knew exactly when, everything each of them had been shown agreed on everything that would matter afterward. One of them saw a thriller and the other a comedy. In the last minutes their films resolved in opposite directions, and on the way out the first thing they asked each other was which one they had seen.

The book has tried to say what makes that evening possible and what makes it worth having.

The apparatus is modest. A projector already shows more flashes than one film needs, and glasses that open on a chosen period and phase can divide those flashes among several films. The division is costly, in light and in flicker, and the costs rise with every stream, so the families that can be shown are small. But the costs fall whenever the films coincide, and the room without glasses sees their average, which can itself be designed.

The form is demanding. A family of films that can share one screening must be written together, against a shared spine, with scenes that meet on a common clock and moments at which every realization agrees on what has happened. Whether a viewer can move from one realization to another is not a matter of taste but of continuity: whether their history can be carried into the other's future without leaving them lost or deceived. Sound, which cannot be divided the way light can, is best composed once for all the realizations, so that the room stays acoustically whole while the pictures diverge.

The stakes are real. A screen that shows different people different things can widen what a film can do: offer comedy and drama, beginner and expert, one ending and another, access and translation, to people who remain together. It can also narrow what a public can know, if the variation is hidden, assigned without consent, sold by tier, locked behind keys in devices the viewers cannot inspect, or used to deliver different messages under one title. The difference between these outcomes is not in the optics. It is in whether the variation is declared, chosen, and made by people with standing to make it.

The central claim can now be stated with the precision the book has earned. Collective viewing does not require that everyone see the same thing. It requires that different experiences stay bounded in time, return repeatedly to agreement, and share at least one channel that everyone receives alike. Where those conditions hold, an audience is neither a mass receiving one message nor a scatter of individuals receiving unrelated ones. It is a plurality that

shares time without sharing every frame.

That arrangement has been present in the idea from its first form: two films about a society whose rulers controlled what everyone saw, made to be watched at once, joined by a score written for both, and separated only by the timing of a shutter each viewer set for themselves. The story was about the danger of one authority deciding what a public sees. The way it was to be shown was an answer to that danger: not one picture for everyone, and not a private picture for each, but several pictures, openly offered, freely chosen, and watched together.

Appendix A

Two-Stream Reference Specification

This appendix specifies a minimal two-stream system, sufficient for a first prototype of the kind described in Appendix E.

Display. A projector or emissive screen refreshing at $R = 144$ slots per second, with a controllable blanking interval of duration β at the start of each slot. Streams occupy period $p = 2$: stream 0 in even slots, stream 1 in odd slots. Each stream receives 72 flashes per second; content captured at 24 images per second is shown three times per image, or content captured at 72 images per second once each.

Gating. Each lens follows the gate $g_{2,\varphi}$ of Chapter 2, opening after the blanking interval of each admitted slot and closing before the next slot begins. With slot duration $1/R \approx 6.94$ ms, the open interval is at most $1/R - \beta$, less the shutter's switching time.

Adaptive gating. When the coincidence map marks a slot as coincident, meaning the displayed images of both streams agree within the declared perceptual tolerance, the glasses remain open through it. Transitions between gated and open operation are scheduled at cuts or ramped over at least several hundred milliseconds, and are never repeated at a regular interval.

Synchronization. A beacon transmits, at least once per frame period, a packet carrying the slot counter, the current coincidence flag for the next several slots, the current protection level, and a schedule identifier. Glasses maintain a local oscillator locked to the slot counter, so that brief loss of the beacon does not lose phase.

Selection. A two-position dial sets $\varphi \in \{0, 1\}$. The glasses store only the dial position and, if the exhibition permits switching under the switch system, the current switch state of Chapter 6.

Failure. On loss of the beacon beyond the oscillator's holdover time, the glasses enter the declared failure mode (transparent, opaque, or common-stream), which the beacon's protection level may specify per passage.

Excluded components. The glasses contain no camera, eye tracker, microphone, general-purpose storage, or transmitter. This exclusion is part of the specification, not an option.

Appendix B

Variant-Manuscript Notation

A variant manuscript records the spine and columns of Chapter 14. Each production unit is written as a block with a shared header and one column per realization. A minimal plain-text form follows.

```
UNIT 14  "The control room"
  ENVELOPE      00:41:10 - 00:43:55
  ANCHORS       00:41:10 door opens (shared sound)
                00:43:55 lights fail (shared sound)
  SPINE         Theseus enters; overrides the broadcast console;
                the dome lights fail.
  COLUMN RNP    Played for dread. Close on Theseus's hands.
                ESTABLISHES console_overridden
                PRESUPPOSES knows_override_code
  COLUMN BNP    Played as a prank. Wide, with the guard asleep.
                ESTABLISHES console_overridden
                PRESUPPOSES knows_override_code
  DIALOGUE      Faces speaking in both columns: "It's done."
                (single recorded line; shared track)
  COINCIDENT    final 4 s (lights fail, both columns)
  RECONCILE     none; rendezvous at end of unit
  PROTECTION    casual
```

The fields correspond to the tuple $\langle A_i, V_i, \Delta_i, Q_i \rangle$: ANCHORS and ENVELOPE to A_i , the COLUMN entries to V_i , ESTABLISHES and PRESUPPOSES to Δ_i , and RECONCILE to Q_i . DIALOGUE, COINCIDENT, and PROTECTION feed the dialogue check, the coincidence map, and the protection profile.

Appendix C

Exhibition Manifest

The manifest of Chapter 24 is published with the variant registry for every screening. A minimal form follows; values in parentheses are placeholders.

WORK	Rarely/Barely Needed Protocols
REGISTRY	(public registry identifier)
SCREENING	(venue, date, start time)
DISPLAY	144 slots/s, period 2, adaptive gating on
SOUND	complementary score, shared
STREAM 0	realization RNP (thriller) phase 0 authors: (credits) fingerprint: (hash)
STREAM 1	realization BNP (satire) phase 1 authors: (credits) fingerprint: (hash)
COMPOSITE	designed; fingerprint: (hash)
SWITCHING	structural rendezvous only
FAILURE MODE	transparent (casual passages), opaque (protected passages)
SPONSORED	none
DISTRIBUTIVE	none

A viewer reads the phase on their own glasses and finds the corresponding stream in the manifest. Nothing about the viewer is recorded.

Appendix D

Evaluation Protocol

The protocol follows the four groups of Chapter 26. Technical tests measure refresh stability, phase drift over a full screening, optical crosstalk, informational legibility of leaked frames at protected moments, luminance through the glasses in coincident and divergent passages, central and peripheral flicker, the timing and regularity of brightness changes, performance across seating positions, and behavior under loss of beacon and power. Narrative tests watch each realization alone, review the family's relations and asymmetries, attempt admissible and forbidden switches including multi-switch paths, and compare the memories of viewers of different realizations. Social tests ask mixed audiences whether they attended one event, how they experienced others' reactions, how the conversation afterward went, and whether they felt ranked. The ethical audit inspects the glasses for excluded components, checks the manifest against the exhibited streams, confirms provenance, and reviews the disclosure given to the audience. A family is not exhibited until all four groups pass.

Appendix E

Prototype Film

A short experimental production would cross three dimensions, each with two options: genre, comedy or drama; explanation, beginner or expert; ending, A or B. The nominal space contains eight realizations. The prototype would not show all eight in one screening, which Part I rules out, but would produce all eight and exhibit them in pairs and triples.

Its purpose is to measure the couplings Chapter 7 predicted. Does changing genre change the time available for explanation? Does each ending remain earned in both genres, or does one ending need different seeds in the comic and dramatic realizations? How far do the eight realizations actually differ in image, fact, and time distance, and how far does the nominal count of eight overstate the family? The answers would show whether dimensions that are analytically independent can be made operationally independent, and at what cost.

Appendix F

Glossary

Adaptive gating Opening the glasses fully while streams coincide on screen, so that multiplexing costs are paid only during divergence.

Blink-rate glasses Glasses whose lenses open and close on a schedule synchronized to the display; the rate belongs to the shutter, not the wearer.

Coincidence Agreement of streams' displayed images within a declared perceptual tolerance.

Common channel Any element of a screening received identically by everyone present.

Composite What the unaided eye sees: the time-weighted average, in linear light, of all streams.

Envelope An interval on the common clock within which every realization must begin and end a scene.

Occasion-preserving family A family exhibited with bounded drift, recurring structural rendezvous, and a common channel.

Period and phase The two integers that fix a periodic gate: how many slots per cycle, and which slot opens.

Protection profile The schedule of leakage requirements across a film.

Realization One complete audiovisual experience within a family.

Rendezvous, structural A time at which every realization establishes the same relevant facts.

Rendezvous, viewer-relative A time at which a particular history admits a switch into every realization.

Selection and concealment Choosing a stream, and preventing someone from seeing one; different functions with different obligations.

Switch admissibility The condition that a history covers what a continuation presupposes and does not contradict what it commits to.

Switch system The transition system whose states summarize viewers' histories for purposes of switching.

Temporal debt A realization's displacement from the spine on the common clock.

Variant family A set of realizations designed for synchronized presentation.

Variant matrix The writing format of spine and columns that records a family's relations.

Appendix G

Provenance

The ideas developed in this book first appear in two conversational notebooks, conversations with a language model archived in the author's public repositories. Only the author's own turns are quoted here, verbatim, including their original spelling. The model's replies elaborated these ideas in various directions, and none of that elaboration is claimed as a source.

Rarely Needed Protocols

Archived at `standardgalactic/library/rarely-needed-protocols.txt` and `standardgalactic/academizer/stack/Rarely Needed Protocols.txt`. Commit dates: to be completed from the repository history.

The frame world:

What if we called black and white televisions "normal televisions" and color TVs "Baird Televisions"?

It's for an alternate world where all televisions are round and square televisions are banned as well as flat screens and only cathode ray tubes or spinning projectors are permitted.

The paired films and the shared soundtrack:

I'm liking it the two versions are *Rarely Needed Protocols* a sci fi thriller, and *Barely Needed Protocols*, which is a satirical comedy show made for kids, but they can be watched simultaneously, as they have overlapping, complementary soundtracks.

The glasses and multiple storylines:

I also wanted to popularize the blink rate glasses needed to watch them as there are alternate plots and story lines set to different blink rates.

Selection by phase:

So they can be manual as well like an old film projector as shot I mean they select different frames like all to 60 fps or something, but it depends which frame you start on

And, after the model's reply interpreted "blink rate" as the viewer's own blinking, the correction that fixes the term's meaning:

I was referring to the shutter mechanism

Transcendental Movie Ideas

Archived at `standardgalactic/mirror/Transcendental Movie Ideas.txt` and `standardgalactic/academizer/stack/Transcendental Movie Ideas.txt`. Commit dates: to be completed from the repository history.

Keyed sequences and protection:

How about blink-rate glasses with patented algorithmic blink rates allowing certain pieces to be seen if they have the correct blink rate algorithm, or blink rate protectors and active lens frames set to the correct blink sequence?

The private viewing room:

I also thought of an art piece that you have to prove you have no cameras or recorders by an airport like scan and are alone in the room where you are shown an original individualized piece of art and given a verbal pass phrase for accessing it online later

Other references

Blink-rate glasses also appear, as an alternative interface and perceptual aid rather than as a cinema device, in `standardgalactic/academizer/stack/Analog Hypermedia.txt` and in several files of `standardgalactic/kitbash`, including a list of alternative interfaces that names "blink-rate glasses with filter controls." A later discussion of blink-rate cinema and multiplexed projection has been reported from another conversation history, dated August 2025; it is noted here as unverified until its transcript is located.

Bibliography

- [1] M. Chion, *Audio-Vision: Sound on Screen*, trans. C. Gorbman (New York: Columbia University Press, 1994).
- [2] *Clue*, dir. J. Lynn (Paramount Pictures, 1985). Released theatrically with three alternative endings distributed to different theaters.
- [3] C. Doctorow, *Information Doesn't Want to Be Free: Laws for the Internet Age* (San Francisco: McSweeney's, 2014).
- [4] U. Eco, *The Open Work*, trans. A. Cancogni (Cambridge, MA: Harvard University Press, 1989); originally *Opera aperta* (1962).
- [5] N. Goodman, *Languages of Art: An Approach to a Theory of Symbols* (Indianapolis: Bobbs-Merrill, 1968).
- [6] C. Gorbman, *Unheard Melodies: Narrative Film Music* (Bloomington: Indiana University Press, 1987).
- [7] L. Hammond, "Stereoscopic motion picture," US Patent 1,435,520, granted November 14, 1922. Publicly exhibited as the Televue system, Selwyn Theatre, New York, December 27, 1922.
- [8] R. Činčera et al., *Kinoautomat: Člověk a jeho dům* (Czechoslovak pavilion, Expo 67, Montreal, 1967).
- [9] *Black Mirror: Bandersnatch*, dir. D. Slade, written by C. Brooker (Netflix, 2018).
- [10] J. Oakes, *Keeping Track: How Schools Structure Inequality* (New Haven: Yale University Press, 1985).
- [11] E. Pariser, *The Filter Bubble: What the Internet Is Hiding from You* (New York: Penguin, 2011).
- [12] Plutarch, *Life of Theseus*, 23.1.
- [13] *Rashomon*, dir. A. Kurosawa (Daiei, 1950).
- [14] Sony Computer Entertainment, announcement of the PlayStation 3D Display with Simul-View, August 2011.

- [15] X. Wu and G. Zhai, "Temporal psychovisual modulation: a new paradigm of information display," *IEEE Signal Processing Magazine* 30(1), 136–141 (2013), doi:10.1109/MSP.2012.2219678.
- [16] S. Zuboff, *The Age of Surveillance Capitalism* (New York: PublicAffairs, 2019).